LINEAR ALGEBRA AND ITS APPLICATIONS

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CHAPTER 1

INTRODUCTION

Linear algebra is a fundamental topic with applications across various fields.

(1) Machine Learning: Algorithms such as linear regression, support vector machines, and principal component analysis, rely on linear algebra concepts.
(2) Signal Processing: Comes up in signal processing applications such as image and audio filtering, detection, and compression.
(3) Control and Networks: Used in control engineering to design control and dynamical systems. Matrices are also used to analyze and model social or transportation networks.
(4) Computer Graphics: Used to represent and manipulate 2D and 3D objects, perform transformations (rotation, scaling, translation), and handle lighting and shading computations.
(5) Operations Research: Linear programming, which comes up in resource allocation and scheduling problems, involves optimizing a linear function subject to linear constraints.
(6) Physical Sciences: Essential in solving problems in classical mechanics, quantum mechanics, and electromagnetism.
(7) Economics and Finance: Used in economic models, financial analysis, and portfolio optimization.

We discuss some concrete problems that can be solved using linear algebra.

(1) **Linear regression**. Given a matrix A and a vector b, find vector x that minimizes

$$\|Ax - b\|^2 = \sum_{i=1}^{m} \Big(\sum_j A_{ij} x_j - b_i \Big)^2,$$

where n is the number of variables and m is the number of measurements. We use this when the equation $Ax = b$ is not solvable and we wish to find an approximate solution. This is the backbone of numerous prediction and inference problems.

(2) **Quadratic optimization**.

More generally, we might have a problem of the form

$$\min_x \; x^\top Q x + q^\top x \quad \text{subject to} \quad Bx = c,$$

where Q is a symmetric matrix and q is a vector.

(3) **Eigenvalue problem**. Given a symmetric matrix Q, we wish to find orthonormal vectors $x_1, \ldots, x_k$ that minimizes
$$x_1^\top Q x_1 + \cdots + x_k^\top Q x_k.$$
For the case $k = 1$, the solution to this problem is the eigenvector with the smallest eigenvalue. However, the solution is far from obvious when $k > 1$. The eigenvalue problem has profound applications ranging from circuit theory and structural engineering to quantum mechanics and control theory.

(4) **Principal component analysis**. Given vectors $z_1, \ldots, z_k$ in an n-dimensional space, find a subspace S of dimension $d < n$ that minimizes
$$\sum_{i=1}^{k} \operatorname{dist}(z_i, S)^2 = \sum_{i=1}^{k} \min_{x \in S} \|x - z_i\|^2.$$
This is not so obvious, but we can transform it into an eigenvalue problem. This has numerous applications, including data compression, noise reduction, clustering, and feature engineering.

(5) **Low-rank approximation**. Given a matrix A of rank k, we wish to find another matrix X of rank $r < k$ that minimizes $\|X - A\|$, where $\|\cdot\|$ is some norm on the space of matrices.

Low-rank approximation has use cases in image and video compression, recommendation systems, feature extraction, and dimensionality reduction.

(6) **Closest orthogonal transform**. Given a matrix A, we wish to find an orthogonal matrix X (used to model rotations or reflections) that is closest to A, namely, that minimizes $\|X - A\|$. This has applications in computer graphics, quantum mechanics, multiview reconstruction, and determination of molecular structure, to name a few.

The above problems involve optimizing an objective function over a feasible (or constraint) set. Two fundamental questions in this regard are:

(i) How do we know beforehand that the problem has a solution?

(ii) Can we develop an efficient algorithm to find the solution?

We will see how these can be addressed using concepts from linear algebra: solution of linear systems, eigendecomposition, singular value decomposition, etc. In fact, the nontrivial part is to develop a mathematical formulation that allows us to reduce them to a linear algebra problem, which then can be solved using efficient algorithms. This comes under the scope of numerical linear algebra and will not be covered in this course.

Since most of you have taken a course on matrices, let's start by discussing some fundamental results that will be covered in this course:

(1) Rank-Nullity theorem.
(2) Diagonalization and Triangularization.
(3) Eigenvalue Decomposition.
(4) Singular Value Decomposition.
(5) Perron-Frobenius theorem.

CHAPTER 2

VECTOR SPACES

A vector space (also called a linear space) is a mathematical abstraction of physical two- and three-dimensional spaces. In three-dimensional space, we can add two vectors and scale a vector, and these operations come with specific properties. By defining these properties axiomatically, we arrive at the abstract concept of a vector space. The advantage of this abstraction is that it extends beyond traditional vectors; the elements of a vector space can be matrices, sequences, or even functions.

To define a vector space, we need the concept of a (scalar) field, which we will denote by $\mathbb{F}$. In this course, $\mathbb{F} = \mathbb{R}$ (real numbers) or $\mathbb{F} = \mathbb{C}$ (complex numbers). We will skip the abstract definition of a field. We just need that $\mathbb{R}$ and $\mathbb{C}$ come with addition and multiplication operations, denoted by $+$ and $*$, and that every non-zero element has a multiplicative inverse.

DEFINITION 1. *A vector space over a field $(\mathbb{F}, +, *)$ is a set $\mathbb{V}$ with operations $+ : \mathbb{V} \times \mathbb{V} \to \mathbb{V}$ and $\cdot : \mathbb{F} \times \mathbb{V} \to \mathbb{V}$ having the following properties.*

(1) $\forall\, u, v \in \mathbb{V} :\ u + v = v + u.$
(2) $\forall\, u, v, w \in \mathbb{F} :\ u + (v + w) = (u + v) + w.$
(3) $\forall\, a, b \in \mathbb{F},\ u \in \mathbb{V} :\ a \cdot (b \cdot u) = (a * b) \cdot u.$
(4) $\forall\, a \in \mathbb{F},\ u, v \in \mathbb{V} :\ a \cdot (u + v) = (a \cdot u) + (b \cdot v).$
(5) $\forall\, a, b \in \mathbb{F},\ u \in \mathbb{V} :\ (a + b) \cdot u = (a \cdot u) + (b \cdot u).$
(6) $\forall\, u \in \mathbb{V} :\ 1_{\mathbb{F}} \cdot u = u.$
(7) $\exists\, 0 \in \mathbb{V} :\ \forall\, u \in \mathbb{V},\ u + 0 = u.$
(8) $\forall\, u \in \mathbb{V},\ \exists\, (-u) \in \mathbb{F} :\ u + (-u) = 0.$

REMARK 1. *Let us note the following.*

(1) *In property 6, $1_{\mathbb{F}}$ denotes the multiplicative identity of $\mathbb{F}$.*
(2) *Property 1 is commutativity, 2 and 3 are associativity, 4 and 5 are distributivity.*
(3) *We call 0 the zero (or null) vector and $-v$ the inverse of v.*
(4) *The same $+$ symbol is used for $\mathbb{F}$ and $\mathbb{V}$ to keep the notations simple.*
(5) *The same $\mathbb{V}$ can give rise to different spaces if we change the $+$ or $\cdot$ operations.*
(6) *We will simply denote $a \cdot u$ as au.*

EXERCISE 1. *Prove that the null vector and the inverse of a given vector are unique.*

EXERCISE 2. *Let $\mathbb{V}$ be a vector space over $\mathbb{F}$. Then show that*

(1) $\forall\, a \in \mathbb{F} :\quad a \cdot 0_{\mathbb{V}} = 0_{\mathbb{V}}.$
(2) $\forall\, v \in \mathbb{V} :\quad 0_{\mathbb{F}} \cdot v = 0_{\mathbb{V}}.$
(3) $\forall\, v \in \mathbb{V} :\quad (-1)_{\mathbb{F}} \cdot v = -v,$

where $0_{\mathbb{F}}$ and $0_{\mathbb{V}}$ are the identity elements of $\mathbb{F}$ and $\mathbb{V}$ and $(-1)_{\mathbb{F}}$ is the inverse of $1_{\mathbb{F}}$.

EXAMPLE 1. *We give some examples of vector spaces.*

(1) $\mathbb{C}$ *is a vector space over* $\mathbb{C}$ *and over* $\mathbb{R}$.
(2) *Let* $\mathbb{F}$ *be a field and* $n \in \mathbb{N}$. *Define*
$$\mathbb{F}^n = \{(x_1, \ldots, x_n) : \ x_1, \ldots, x_n \in \mathbb{F}\}.$$
Define $+$ *and* $\cdot$ *as follows:*
$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n),$$
and
$$\forall\, a \in \mathbb{F} : \quad a \cdot (x_1, \ldots, x_n) = (ax_1, \ldots, ax_n).$$
Then $(\mathbb{F}^n, +, \cdot)$ *is a vector space over* $\mathbb{F}$.
(3) *Let* X *be a set and* $\mathbb{F} = (\mathbb{F}, +, *)$ *be a field. Let* $\mathcal{F}$ *be the collection of functions* $f : X \to \mathbb{F}$. *Then* $\mathcal{F}$ *is a vector space with the following natural operations:*
$$\forall x \in X : \quad (f + g)(x) = f(x) + g(x),$$
and
$$\forall x \in X : \quad (a \cdot f)(x) = a * f(x).$$

EXERCISE 3. *Let* $\mathcal{S}$ *be the collection of complex-valued sequences* $x = (x_1, x_2, \ldots)$, *where* $x_n \in \mathbb{C}$ *for all* $n \in \mathbb{N}$. *Let*
$$\mathcal{L}^1 = \{x \in \mathcal{S} : \ |x_1| + |x_2| + \cdots < \infty\},$$
and
$$\mathcal{L}^2 = \{x \in \mathcal{S} : \ |x_1|^2 + |x_2|^2 + \cdots < \infty\}.$$
Verify that $\mathcal{L}^1$ *and* $\mathcal{L}^2$ *are vector spaces over* $\mathbb{C}$.

EXERCISE 4. *Let* $\mathbb{GL}_n(\mathbb{C})$ *be the set of* $(n \times n)$ *complex-valued matrices that are invertible. Is* $\mathbb{GL}_n(\mathbb{C})$ *a vector space over* $\mathbb{C}$?

EXERCISE 5. *Verify that the set of continuous functions* $f : \mathbb{R} \to \mathbb{R}$ *form a vector space over* $\mathbb{R}$.

The example in Exercise 5 demonstrates that thinking of a vector as merely a point in a coordinate space can be misleading.

A recurring theme in linear algebra is to decompose a vector space into smaller vector spaces with specific structures, known as subspaces.

DEFINITION 2. *Let* $(\mathbb{V}, +, \cdot)$ *be a vector space over* $\mathbb{F}$. *A nonempty set* $\mathbb{U} \subseteq \mathbb{V}$ *is called a subspace of* $\mathbb{V}$ *if* $(\mathbb{U}, +, \cdot)$ *is a vector space over* $\mathbb{F}$.

REMARK 2.

(1) *When we say* $\mathbb{U}$ *is a subspace, it is understood that* $\mathbb{U}$ *is a subset of some vector space.*
(2) *In some textbooks, instead of asking* $\mathbb{U}$ *to be nonempty, the condition* $0 \in \mathbb{U}$ *is used; these are in fact equivalent.*
(3) *Once we fix* $\mathbb{U} \subset \mathbb{V}$, *we must have* $u + v \in \mathbb{V}$ *and* $au \in \mathbb{V}$ *for all* $u, v \in \mathbb{U}$ *and* $a \in \mathbb{F}$; *what is not obvious is that* $u + v \in \mathbb{U}$ *and* $au \in \mathbb{U}$.

PROPOSITION 1. *A nonempty set* $\mathbb{U} \subseteq \mathbb{V}$ *is a subspace of* $\mathbb{V}$ *if and only if* $u + v \in \mathbb{U}$ *and* $au \in \mathbb{U}$ *for all* $u, v \in \mathbb{U}$ *and* $a \in \mathbb{F}$.

One direction is obvious. In the other direction, all we need to check is that $0 \in \mathbb{V}$ also belongs to $\mathbb{U}$ and that for all $u \in \mathbb{U}$, we have $-u \in \mathbb{U}$; the rest of the vector space axioms hold automatically.

EXAMPLE 2. *In the examples below, $\mathbb{V}$ is a vector space and $\mathbb{U} \subset \mathbb{V}$ is a subspace.*

(1) $\mathbb{V} = \mathbb{F}^n$ *and* $\mathbb{U} = \{x \in \mathbb{V} : \ x_1 = 0\}$.
(2) $\mathbb{V} = \mathbb{F}^n$ *and* $\mathbb{U} = \{x \in \mathbb{V} : \ x_1 = \cdots = x_n\}$.
(3) $\mathbb{V} = \mathbb{F}^n$ *and* $\mathbb{U} = \{x \in \mathbb{V} : \ a_1 x_1 + \cdots + a_n x_n = 0\}$ *where* $a \in \mathbb{F}^n$.
(4) $\mathbb{V}$ *is the space of polynomials and* $\mathbb{U} = \{p \in \mathbb{V} : \ p(0) = 0\}$.

EXERCISE 6. *Let $\mathbb{V}$ be the space of complex-valued sequences and let $\mathbb{U}$ consist of those $x \in \mathbb{V}$ for which $x_n \to 0$ as $n \to \infty$. Verify that $\mathbb{U}$ is a subspace of $\mathbb{V}$.*

($x_n \to 0$ means that, for all $\epsilon > 0$, there exists $N \geqslant 1$, such that $|x_n| < \epsilon$ for all $n \geqslant N$.)

EXERCISE 7. *Let $a = (a_1, \ldots, a_n) \in \mathbb{R}^n$. Show that*

$$\mathcal{H} = \big\{(x_1, \ldots, x_n) \in \mathbb{R}^n : \ a_1 x_1 + \cdots + a_n x_n = 0\big\}$$

is a subspace of $\mathbb{R}^n$.

EXERCISE 8. *Let $\mathbb{V}$ be a vector space over $\mathbb{F}$. Let X be a nonempty subset of $\mathbb{V}$ with the property that*

$$\forall\, u, v \in X,\ t \in \mathbb{F} : \quad tu + (1-t)v \in X.$$

Prove that there exists an $x_0 \in X$ and a subspace $\mathbb{U} \subseteq \mathbb{V}$ such that $X = x_0 + \mathbb{U}$.

We now describe two commonly used operations for generating new subspaces from existing ones.

PROPOSITION 2. *Let $(\mathbb{U}_\alpha)_{\alpha \in I}$ be a collection of subspaces of $\mathbb{V}$, where I is some index set. Then*

$$\bigcap_{\alpha \in I} \mathbb{U}_\alpha \ \text{ is a subspace of } \ \mathbb{V}.$$

Unfortunately, the union of two subspaces need not be a subspace. For example, let if $\mathbb{V} = \mathbb{R}^2$, $\mathbb{U}_1 = \{x \in \mathbb{V} : \ x_1 = 0\}$, and $\mathbb{U}_2 = \{x \in \mathbb{V} : \ x_2 = 0\}$, then $\mathbb{U}_1 \cup \mathbb{U}_2$ is a cross-shaped object which is not closed under addition.

DEFINITION 3. *Let $\mathbb{U}_1, \ldots, \mathbb{U}_m$ be subspaces of $\mathbb{V}$. Their sum is defined as*

$$\mathbb{U}_1 + \cdots + \mathbb{U}_m = \big\{u_1 + \cdots + u_m : \ u_1 \in \mathbb{U}_1, \ldots, u_m \in \mathbb{U}_m\big\}.$$

EXAMPLE 3. *Consider the example above: $\mathbb{U}_1 = \{x \in \mathbb{V} : \ x_1 = 0\}$ and $\mathbb{U}_2 = \{x \in \mathbb{V} : x_2 = 0\}$. Then $U_1 + U_2 = \mathbb{R}^2$.*

The sum of subspaces is the smallest subspace containing their union.

EXERCISE 9. *Verify that $\mathbb{U}_1 + \cdots + \mathbb{U}_m$ is subspace of $\mathbb{V}$. Moreover, show that it is the smallest subspace containing $\mathbb{U}_1, \ldots, \mathbb{U}_m$, namely, for any subspace $\mathbb{W} \subset \mathbb{V}$ containing $\mathbb{U}_1 \cup \ldots \cup \mathbb{U}_m$, we have $\mathbb{U}_1 + \cdots + \mathbb{U}_m \subseteq \mathbb{W}$.*

We again revisit the example $\mathbb{U}_1 = \{x \in \mathbb{V} : \ x_1 = 0\}$ and $\mathbb{U}_2 = \{x \in \mathbb{V} : \ x_2 = 0\}$. For every $v \in \mathbb{U}_1 + \mathbb{U}_2 = \mathbb{R}^2$, there exists $u_1 \in \mathbb{U}_2$ and $u_2 \in \mathbb{U}_2$ such that $v = u_1 + u_2$. This decomposition is unique and is called a "direct" sum.

DEFINITION 4. *Let $\mathbb{U}_1, \dots, \mathbb{U}_m$ be subspaces of $\mathbb{V}$. We say that $\mathbb{U}_1 + \cdots + \mathbb{U}_m$ is a direct sum if every $v \in \mathbb{U}_1 + \cdots + \mathbb{U}_m$ can be uniquely written as $v = u_1 + \cdots + u_m$ form some $u_i \in \mathbb{U}_i$. We use the symbol $\mathbb{U}_1 \oplus \cdots \oplus \mathbb{U}_m$ for a direct sum.*

We give a straightforward characterization of the direct sum for $m = 2$.

EXERCISE 10. *Prove that $\mathbb{U}_1 + \cdots + \mathbb{U}_m$ is a direct sum if and only if $u_1 + \cdots + u_m = 0$ and $u_i \in \mathbb{U}_i$ implies $u_i = 0$. In particular, show that $\mathbb{U}_1 + \mathbb{U}_2$ is a direct sum if and only if $\mathbb{U}_1 \cap \mathbb{U}_2 = 0$.*

Next, we look at the concepts of span, independence, and basis. All of these rest on the idea of a "linear combination," which in turn is built on the two operational pillars of a vector space.

DEFINITION 5. *Let $v_1, \dots, v_m \in \mathbb{V}$. We say that $v \in \mathbb{V}$ is a linear combination of $v_1, \dots, v_m$ if there exists $a_1, \dots a_m \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_m v_m$. The set of all possible linear combinations is called the span of $v_1, \dots, v_m$:*

$$\operatorname{span}(v_1, \dots, v_m) = \left\{ a_1 v_1 + \cdots + a_m v_m : \ a_1, \dots, a_m \in \mathbb{F} \right\}.$$

EXAMPLE 4.

(1) *Any $v \in \mathbb{R}^2$ is in the span of $v_1 = (1, 0)$ and $v_2 = (0, 1)$.*
(2) *The vector $v = (1, 1, 1)$ is not in the span of $v_1 = (1, 0, 0)$ and $v_2 = (0, 1, 0)$.*

EXERCISE 11. *Let $v_1 = (1, -1)$ and $v_2 = (-1, 1)$. Show that $\operatorname{span}(v_1, v_2) = \mathbb{R}^2$.*

EXERCISE 12. *Show that $\mathbb{U} = \operatorname{span}(v_1, \dots, v_m)$ is a subspace of $\mathbb{V}$. Moreover, argue why $\mathbb{U}$ is the smallest subspace containing $v_1, \dots, v_m$.*

We now define the core concept for this course, the notion of finite-dimensional vector spaces.

DEFINITION 6. *A vector space $\mathbb{V}$ is finite-dimensional if there exists $n \geqslant 1$ and $v_1, \dots, v_n \in \mathbb{V}$ such that $\mathbb{V} = \operatorname{span}(v_1, \dots, v_n)$. A vector space that is not finite-dimensional is called infinite-dimensional.*

Note that we have not defined "dimension" at this point. Moreover, n in the above definition is not necessarily the dimension of the vector space.

EXAMPLE 5.

(1) *$\mathbb{F}^m$ is finite-dimensional.*
(2) *The space $\mathbb{F}^{m \times n}$ of $(m \times n)$ $\mathbb{F}$-valued matrices is finite-dimensional.*
(3) *The space of polynomials of degree at most k is finite-dimensional.*

EXERCISE 13. *Prove that every subspace of a finite-dimensional vector space is finite-dimensional.*

EXERCISE 14. *Verify that $\mathbb{R}$ is a vector space over $\mathbb{R}$. Also, verify that $\mathbb{R}$ is a vector space over $\mathbb{Q}$. Is $\mathbb{R}$ finite-dimensional over $\mathbb{R}$? Is $\mathbb{R}$ finite-dimensional over $\mathbb{Q}$?*

EXERCISE 15. *Show that the following spaces are infinite-dimensional.*

(1) *The space of polynomials with complex coefficients.*
(2) *The space of continuous functions $f : \mathbb{R} \to \mathbb{R}$.*

EXERCISE 16. *Let $\mathbb{V}$ be a vector space. Show that the following are equivalent:*

(1) *$\mathbb{V}$ is infinite-dimensional.*
(2) *For all $k \geqslant 1$, there exists $v_1, \ldots, v_k \in \mathbb{V}$ that are linearly independent.*
(3) *For all $k \geqslant 1$ and $v_1, \ldots, v_k \in \mathbb{V}$, there exists $v \in \mathbb{V}$ that does not belong to $\operatorname{span}(v_1, \ldots, v_k)$.*

DEFINITION 7. *We say that $v_1, \ldots, v_m \in \mathbb{V}$ are linearly independent if*

$$a_1 v_1 + \ldots + a_m v_m = 0 \Rightarrow a_1 = \ldots = a_m = 0.$$

In other words, there does not exist $a_1, \ldots, a_m$ that are not all zero and yet

$$a_1 v_1 + \ldots + a_m v_m = 0.$$

If the vectors are not linearly independent, we say they are linearly dependent.

EXERCISE 17.

(1) *Show that $v_1 = (1, -1)$ and $v_2 = (-1, 1)$ are linearly independent in $\mathbb{R}^2$.*
(2) *In $\mathbb{F}^n$, define vectors $e_1, \ldots, e_n$ to be*

$$\forall\, i = 1, \ldots, n,\ \forall\, j = 1, \ldots, n: \quad e_i(j) = \begin{cases} 1, & \text{if } j = i, \\ 0, & \text{if } j \neq i. \end{cases}$$

Show that $e_1, \ldots, e_n$ are linearly independent. We call this the standard basis.
(3) *Show that the system of functions $e_1, e_2, \ldots : [0, 1] \to \mathbb{C}$ defined as*

$$e_k(t) = \exp(\iota 2\pi t) \qquad (t \in [0, 1]),$$

are linearly independent in the space of complex-valued functions on $[0, 1]$.

EXERCISE 18. *If $v_1, \ldots, v_m$ are linearly independent, then show that*

$$\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(v_1) \oplus \cdots \oplus \operatorname{span}(v_m).$$

We can generalize the above result as follows.

PROPOSITION 3. *Let $\mathbb{V} = \operatorname{span}(v_1, \ldots, v_m)$ and let $u_1, \ldots, u_n \in \mathbb{V}$ be linearly independent. Then $m \geqslant n$.*

Thus, a clear relationship exists between the cardinality of spanning vectors and linearly independent vectors. As a consequence of the above observation, we can conclude the following.

COROLLARY 1. *Any $(n + 1)$ vectors in $\mathbb{F}^n$ must be linearly dependent and any $(n - 1)$ vectors cannot span $\mathbb{F}^n$.*

We can prove Proposition 3 using the following lemma.

LEMMA 1. *Let $v_1, \ldots, v_k$ be linearly independent and let $v_{k+1}, \ldots, v_{k+p}$ be such that $v_1, \ldots, v_{k+p}$ are linearly dependent. Then there exists $1 \leqslant j \leqslant p$ such that*

$$v_{k+j} \in \operatorname{span}(v_1, \ldots, v_{k+j-1}).$$

DEFINITION 8. *A basis for $\mathbb{V}$ is a set of linearly independent vectors that span $\mathbb{V}$.*

EXERCISE 19.

(1) *Show that $e_1, \ldots, e_n$ in Exercise 17 are linearly independent.*
(2) *Find a basis for the space of matrices $\mathbb{F}^{m\times n}$.*
(3) *Find a basis for the space of cubic polynomials.*

In fact, we can always find a basis if the space is finite-dimensional.

THEOREM 1. *Every finite-dimensional vector space has a basis.*

Later, we will see that every finite-dimensional vector space has a special basis known as an orthonormal basis, which is easier to work with.

We next give a useful characterization of a basis (this is the definition of a basis in some books).

EXERCISE 20. *Show that a set of vectors $\{v_1, \ldots, v_n\}$ is a basis of $\mathbb{V}$ if and only if, for all $v \in \mathbb{V}$, there exists unique $a_1, \ldots, a_n \in \mathbb{F}$ such that $v = a_1 v_1 + \ldots + a_n v_n$.*

DEFINITION 9. *For a finite-dimensional vector space $\mathbb{V}$, we define its dimension to be the cardinality of a basis of $\mathbb{V}$. We denote this by $\dim \mathbb{V}$.*

The trivial space $\mathbb{V} = \{0\}$ does not admit a basis; its dimension is defined as 0.

The following result confirms that the notion of "dimension" does not depend on the choice of basis; it is an intrinsic quantity for a vector space.

PROPOSITION 4. *Let $\mathbb{V}$ be a finite-dimensional vector space. Then any two bases of $\mathbb{V}$ have the same cardinality.*

EXERCISE 21. *Let $\dim \mathbb{V} = n$. Show that*

(1) *Any n linearly independent vectors is a basis of $\mathbb{V}$.*
(2) *Any n vectors that span $\mathbb{V}$ is a basis of $\mathbb{V}$.*

EXERCISE 22. *Let $\mathbb{V}$ be a finite-dimensional space and $\mathbb{U}$ be a subspace of $\mathbb{V}$. Then show that $\dim \mathbb{U} \leqslant \dim \mathbb{V}$.*

EXERCISE 23. *Find $\dim \mathcal{H}$ where $\mathcal{H}$ is the subspace in Exercise 7.*

EXERCISE 24. *Let $\mathbb{U}_1$ and $\mathbb{U}_2$ be two subspaces of a vector space $\mathbb{V}$. If $\dim \mathbb{U}_1 + \dim \mathbb{U}_2 > \dim \mathbb{V}$, then prove that there must exist a nonzero vector in $\mathbb{U}_1 \cap \mathbb{U}_2$.*

EXERCISE 25. *Let $\mathbb{V}$ be a vector space over $\mathbb{R}$. A subspace $\mathbb{U} \subseteq \mathbb{V}$ is proper if $\mathbb{U} \neq \mathbb{V}$. Prove that if $\dim \mathbb{V} \geqslant 1$, then $\mathbb{V}$ cannot be the union of proper subspaces.*

Next, we introduce the concept of a complementary subspace.

PROPOSITION 5. *Let $\mathbb{V}$ be a finite-dimensional space and $\mathbb{U}$ be a subspace of $\mathbb{V}$. Then there exists a subspace $\mathbb{W}$ of $\mathbb{V}$ such that $\mathbb{V} = \mathbb{U} \oplus \mathbb{W}$.*

We call $\mathbb{W}$ the complementary subspace of $\mathbb{U}$. Note that the complementary subspace is not unique.

EXERCISE 26. *In Propsotion 5, prove that $\dim \mathbb{V} = \dim \mathbb{U} + \dim \mathbb{W}$.*

EXERCISE 27. *Let $\mathbb{V} = \mathbb{R}^2$ and $\mathbb{U} = \{(x, 0) : x \in \mathbb{R}\}$. Show that*

$$\mathbb{W}_1 = \{(0, x) : x \in \mathbb{R}\} \quad \textit{and} \quad \mathbb{W}_2 = \{(x, x) : x \in \mathbb{R}\}$$

are complementary subspaces of $\mathbb{U}$.

CHAPTER 3

LINEAR TRANSFORMS AND MATRICES

We now look at the central objects of study in linear algebra — linear transforms and matrices. We recall that the underlying field $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

DEFINITION 10. *Let $\mathbb{V}$ and $\mathbb{W}$ be two vector spaces over $\mathbb{F}$. A linear transform from $\mathbb{V}$ to $\mathbb{W}$ is a function $T : \mathbb{V} \to \mathbb{W}$ with the following properties:*

$$\begin{aligned} \forall\, u, v \in \mathbb{V} : \quad & T(u+v) = T(u) + T(v). \\ \forall\, u \in \mathbb{V},\, a \in \mathbb{F} : \quad & T(au) = aT(u). \end{aligned}$$

REMARK 3.

(1) *Linear transforms are also called linear maps or linear operators. In this course, we will reserve the term "operator" for the case where $\mathbb{V} = \mathbb{W}$.*
(2) *We will use $\mathcal{L}(\mathbb{V}, \mathbb{W})$ to denote the set of all possible linear transforms from $\mathbb{V}$ to $\mathbb{W}$. In particular, we will use the shorthand $\mathcal{L}(\mathbb{V}) = \mathcal{L}(\mathbb{V}, \mathbb{V})$.*
(3) *A linear transform $T : \mathbb{V} \to \mathbb{W}$ maps $0_{\mathbb{V}}$ to $0_{\mathbb{W}}$.*
(4) *We will use 0 and Id to denote the zero and identity maps:*

$$\forall\, v \in \mathbb{V} : \quad 0(v) = 0 \quad \textit{and} \quad \mathrm{Id}(v) = v.$$

EXAMPLE 6.

(1) *$\mathbb{V} = \mathbb{W} = \mathbb{R}$, and $T(x) = ax$, where $a \in \mathbb{R}$.*
(2) *$\mathbb{V} = \mathbb{R}^n$, $\mathbb{W} = \mathbb{R}$, and $T(x) = a_1x_1 + a_2x_2$, where $a_1, a_2 \in \mathbb{R}$.*
(3) *$\mathbb{V} = \mathbb{W} = \mathbb{R}^2$, and $T(x_1, x_2) = (x_2, x_1)$.*
(4) *$\mathbb{V}$ and $\mathbb{W}$ are the space of polynomials and T is the derivative operator.*
(5) *$\mathbb{V}$ is the space of continuous functions on $[0, 1]$, $\mathbb{W} = \mathbb{R}$, and*

$$\forall\, f \in C([0,1]) : \quad T(f) = \int_0^1 f(t)\, dt.$$

Linear transforms come up naturally in defining a "coordinate map" associated with a finite-dimensional vector space.

DEFINITION 11. *Let $\mathbb{V}$ be finite-dimensional vector space and $\dim \mathbb{V} = n$. Fix a basis $\{v_1, \ldots, v_n\}$ of $\mathbb{V}$. Define*

$$\phi : \mathbb{V} \to \mathbb{F}^n, \quad \phi(v) = (x_1, \ldots, x_n),$$

where $v = x_1v_1 + \cdots + x_nv_n$ is the unique representation of v in the basis. We call ϕ the coordinate map and $x = \phi(v)$ the coordinates of v.

In fact, ϕ has more structure to it than just linearity — it is an isomorphism.

DEFINITION 12. *A linear map $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ is called an isomorphism if it is a bijection, i.e., if for all $w \in \mathbb{W}$, there exists a unique $v \in \mathbb{V}$ such that $w = T(v)$. We say that $\mathbb{V}$ and $\mathbb{W}$ are isomorphic (written as $\mathbb{V} \cong \mathbb{W}$) if there exists an isomorphism $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$.*

EXERCISE 28. *Show that the coordinate map is a linear isomorphism from $\mathbb{V}$ to $\mathbb{F}^{\dim \mathbb{V}}$.*

Every isomorphism admits an "inverse", which is again linear.

EXERCISE 29. *Let $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ be an isomorphism. Show that $T^{-1} \in \mathcal{L}(\mathbb{W}, \mathbb{V})$ is also an isomorphism.*

The set $\mathcal{L}(\mathbb{V}, \mathbb{W})$ comes with a natural vector space structure. Namely, if $T, T' \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ and $a \in \mathbb{F}$, then we can define $T + T' : \mathbb{V} \to \mathbb{W}$ and $aT : \mathbb{V} \to \mathbb{W}$ as follows:

$$\forall\, v \in \mathbb{V}: \quad (T + T')(v) = T(v) + T'(v) \quad \text{and} \quad (aT)(v) = aT(v).$$

EXERCISE 30. *Verify that $T + T'$ and aT are linear transforms and that $\mathcal{L}(\mathbb{V}, \mathbb{W})$ is a vector space over $\mathbb{F}$. In particular, prove that $\dim \mathcal{L}(\mathbb{V}, \mathbb{W}) = \dim \mathbb{V} \cdot \dim \mathbb{W}$.*

Next, we introduce two natural subspaces associated with a linear transform.

DEFINITION 13. *Let $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$. The null space and range space of T are defined to be*

$$\mathcal{N}(T) = \big\{v \in \mathbb{V} : \ T(v) = 0\big\} \quad \textit{and} \quad \mathcal{R}(T) = \{T(v) : \ v \in \mathbb{V}\}.$$

REMARK 4.

(1) *$\mathcal{N}(T)$ and $\mathcal{R}(T)$ are nonempty as $0_{\mathbb{V}} \in \mathcal{N}(T)$ and $0_{\mathbb{W}} \in \mathcal{R}(T)$.*
(2) *$\mathcal{N}(T) \subseteq \mathbb{V}$ and $\mathcal{R}(T) \subseteq \mathbb{W}$.*
(3) *If $T = 0$, then $\mathcal{N}(T) = \mathbb{V}$ and $\mathcal{R}(T) = \{0\}$.*
(4) *If $T = \mathrm{Id}$, then $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T) = \mathbb{W}$.*

EXERCISE 31. *Let $\mathbb{V} = \mathbb{R}^2$, $\mathbb{W} = \mathbb{R}$, and $T(x) = x_1 + x_2$. Find $\mathcal{N}(T)$ and $\mathcal{R}(T)$ and give their geometric descriptions.*

Since T is linear, we can say more about $\mathcal{N}(T)$ and $\mathcal{R}(T)$.

EXERCISE 32. *Show that $\mathcal{N}(T)$ is a subspace of $\mathbb{V}$ and $\mathcal{R}(T)$ is a subspace of $\mathbb{W}$.*

Since $\mathcal{N}(T)$ and $\mathcal{R}(T)$ are subspaces, we have the following definitions.

DEFINITION 14. *The nullity and rank of T are defined as*

$$\operatorname{nullity} T = \dim \mathcal{N}(T) \quad \textit{and} \quad \operatorname{rank} T = \dim \mathcal{R}(T).$$

EXERCISE 33. *Find* $\operatorname{nullity} T$ *and* $\operatorname{rank} T$ *for the example in Exercise 31.*

There exists a precise relation between these quantities.

THEOREM 2. *Let $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$. If $\mathbb{V}$ is finite-dimensional, then $\mathcal{R}(T)$ is finite-dimensional, and*

$$\operatorname{nullity} T + \operatorname{rank} T = \dim \mathbb{V}.$$

REMARK 5. *We will refer to this result as the rank-nullity theorem. For the above result to hold, $\mathbb{V}$ must be finite-dimensional, but $\mathbb{W}$ can be infinite-dimensional.*

As a direct corollary of rank-nullity, we have the following results.

COROLLARY 2. *For all* $T \in \mathcal{L}(\mathbb{V})$, $\mathcal{N}(T) = \{0\} \iff \mathcal{R}(T) = \mathbb{V}$.

COROLLARY 3. *Let* $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$. *If* $\dim \mathbb{W} < \dim \mathbb{V}$, *then* T *cannot be injective. On the other hand,* $\dim \mathbb{W} > \dim \mathbb{V}$, *then* T *cannot be surjective.*

We next discuss an important class of operators. For any $P \in \mathcal{L}(\mathbb{V})$, we define $P^2 = P \circ P$.

DEFINITION 15. *We say that* $P \in \mathcal{L}(\mathbb{V})$ *is a projection operator (or projector) if* $P^2 = P$.

EXAMPLE 7. *We give some examples of projection operators below.*

(1) *Zero and identity maps.*
(2) $\mathbb{V} = \mathbb{R}^2$, $P \in \mathcal{L}(\mathbb{V}) : \; P(x_1, x_2) = (x_1, 0)$.
(3) $\mathbb{V} = \mathbb{R}^2$, $P \in \mathcal{L}(\mathbb{V}) : \; P(x) = (x_1 + x_2, 0)$.

EXERCISE 34. *Show that* P *is a projector if and only if* $\mathrm{Id} - P$ *is a projector.*

The relation between projection and direct sum is described next.

THEOREM 3. *Let* $P \in \mathcal{L}(\mathbb{V})$ *be a projection operator. Then we can decompose* $\mathbb{V}$ *as* $\mathbb{V} = \mathcal{N}(P) \oplus \mathcal{R}(P)$. *Conversely, if* $\mathbb{V} = \mathbb{U} \oplus \mathbb{W}$, *then we can find a projection operator* $P \in \mathcal{L}(\mathbb{V})$ *such that* $\mathbb{U} = \mathcal{N}(P)$ *and* $\mathbb{W} = \mathcal{R}(P)$.

In the latter case, we call P the projection of $\mathbb{V}$ onto $\mathbb{U}$ along $\mathbb{W}$.

We conclude this part by connecting linear transforms and matrices. A matrix is just an array of numbers. More precisely, a $\mathbb{F}$-valued matrix A of size $(m \times n)$ is represented as

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ & \vdots & \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \qquad (a_{ij} \in \mathbb{F}). \tag{1}$$

We say that A has m rows and n columns. The (i, j)-th component of A is a_{ij}. For $1 \leqslant j \leqslant n$, the vector

$$c_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} \in \mathbb{F}^{m \times 1}$$

is the j-th column of A. Similarly, for $1 \leqslant i \leqslant m$, the vector

$$r_i = (a_{i1} \cdots a_{in}) \in \mathbb{F}^{1 \times n}$$

is the i-th row of A.

Matrices can be viewed as linear transforms in their own right. The space of $\mathbb{F}$-valued $(m \times n)$ matrices is denoted by $\mathbb{F}^{m \times n}$.

DEFINITION 16. *Let* $A \in \mathbb{F}^{m \times n}$ *and* $B \in \mathbb{F}^{n \times p}$. *Their product is the matrix* $C \in \mathbb{F}^{m \times p}$ *given by*

$$\forall\, 1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant p : \quad c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}.$$

We denote this as $C = AB$.

EXERCISE 35. *Let* $m, n \geqslant 1$ *and* A *be defined as in* (1). *Consider the map* $T : \mathbb{F}^n \to \mathbb{F}^m$ *defined as*

$$\forall\, x \in \mathbb{F}^n : \quad (T(x))_i = \sum_{j=1}^{n} a_{ij}\, x_j \qquad (1 \leqslant i \leqslant m).$$

Show that $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$.

Any linear transform between finite-dimensional spaces can be represented using matrices. This is similar to using a coordinate map to represent a vector in a vector space.

EXERCISE 36. *Let* $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$, *where* $\dim \mathbb{V} = n$ *and* $\dim \mathbb{W} = m$. *Let* $v_1, \ldots, v_n$ *be a basis of* $\mathbb{V}$ *and* $\phi : \mathbb{V} \to \mathbb{F}^n$ *be the corresponding coordinate map. Similarly, let* $w_1, \ldots, w_m$ *be a basis of* $\mathbb{W}$ *and* $\psi : \mathbb{W} \to \mathbb{F}^m$ *be the corresponding coordinate map.*

For all $1 \leqslant j \leqslant n$, *there exists* $a_{1j}, \ldots, a_{mj} \in \mathbb{F}$ *such that*

$$T(v_j) = \sum_{i=1}^{m} a_{ij}\, w_i.$$

Define $A \in \mathbb{F}^{m \times n}$ *as in* (1). *Show that* $T = \psi^{-1} \circ A \circ \phi$.

We call A the "matrix representation" of T and denote it as $A = [T]$.

REMARK 6.

(1) *We obtain different matrix representations using different sets of bases, i.e., a matrix representation depends on the choice of bases.*
(2) *We will exploit this idea when we discuss different matrix decompositions. In particular, we will ask whether a given transform can be "diagonalized" using appropriate bases. A diagonal representation is useful for computational reasons.*

EXERCISE 37. *Let* $T_1 \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ *and* $T_2 \in \mathcal{L}(\mathbb{U}, \mathbb{V})$. *For a fixed choice of bases for* $\mathbb{U}, \mathbb{V}$ *and* $\mathbb{W}$, *show that*

$$[T_1 \circ T_2] = [T_1][T_2].$$

EXERCISE 38. *Let* $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ *be given by* $T(x_1, x_2) = (x_1 + 2x_2, 2x_1 + 4x_2)$. *Find* $[T]$ *for the following choice of bases:*

(1) $v_1 = (1, 0)$ *and* $v_2 = (0, 1)$; $w_1 = (1, 0)$ *and* $w_2 = (0, 1)$.
(2) $v_1 = (1, 2)$ *and* $v_2 = (-2, 1)$; $w_1 = (1, 0)$ *and* $w_2 = (0, 1)$.
(3) $v_1 = (1, 2)$ *and* $v_2 = (-2, 1)$; $w_1 = (1, 2)$ *and* $w_2 = (-2, 1)$.

What is the relation between the different matrix representations in Exercise 38.

DEFINITION 17. *We say that* $M, N \in \mathbb{F}^{n \times n}$ *are similar, denoted* $M \sim N$, *if there exists an invertible matrix* $P \in \mathbb{F}^{n \times n}$ *such that* $M = PNP^{-1}$.

EXERCISE 39. *Let* M *and* N *be matrix representations of a linear operator corresponding to two different bases. Prove that* $M \sim N$.

In Problem (14), we show that the space of matrices $\mathbb{F}^{n \times n}$ can be partitioned into equivalence classes, where each equivalence class $[M] = \{N : N \sim M\}$ consists of matrix representations of the linear transform represented by M.

CHAPTER 4

INNER-PRODUCT AND NORMED SPACES

We first recall the concepts of dot product and Euclidean distance on $\mathbb{R}^n$ and $\mathbb{C}^n$. The dot product is defined as

$$x \cdot y = x_1 y_1 + \cdots + x_n y_n \qquad (x, y \in \mathbb{R}^n), \tag{2}$$

and

$$x \cdot y = \bar{x}_1 y_1 + \cdots + \bar{x}_n y_n \qquad (x, y \in \mathbb{C}^n), \tag{3}$$

where $\bar{z}$ denotes the conjugate of $z \in \mathbb{C}$. The length of $x \in \mathbb{R}^n$ is given by

$$\|x\| = \sqrt{x \cdot x} = (x_1^2 + \cdots + x_n^2)^{\frac{1}{2}}, \tag{4}$$

and that of $z \in \mathbb{C}^n$ is

$$\|z\| = \sqrt{z \cdot z} = (|z_1|^2 + \cdots + |z_n|^2)^{\frac{1}{2}}. \tag{5}$$

The Euclidean distance between x and y in $\mathbb{R}^n$ or $\mathbb{C}^n$ is defined as $\|x - y\|$.

The above structures form the basis of Euclidean geometry. Can we extend these definitions to the space of polynomials or continuous functions on $[0, 1]$? More generally, how do we define them for an abstract vector space? This brings us to the concepts of "inner product" and "norm". We will see that while every inner product gives rise to a norm, the converse is not true.

To define the inner product, we must distinguish between real and complex spaces. We call $\mathbb{V}$ a real (resp. complex) vector space if $\mathbb{F} = \mathbb{R}$ (resp. $\mathbb{F} = \mathbb{C}$).

DEFINITION 18. *Let $\mathbb{V}$ be real vector space. An inner product on V is a map $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$ such that*

(1) $\langle x, x \rangle \geqslant 0$ *for all* $x \in \mathbb{V}$, *and* $\langle x, x \rangle = 0$ *if and only if* $x = 0$.
(2) $\forall\, x, y \in \mathbb{V} : \ \langle x, y \rangle = \langle y, x \rangle$.
(3) $\forall\, y \in \mathbb{V}$, *the map* $x \mapsto \langle y, x \rangle$ *is linear.*

REMARK 7. *Property* 1 *is called positive definiteness and* 2 *as symmetry. It follows from properties* 1-3 *that the map* $x \mapsto \langle x, y \rangle$ *is linear for all* $y \in \mathbb{V}$. *For this reason, we refer to* 3 *as bilinearity. Thus, any positive definite, symmetric, bilinear form on* $\mathbb{V}$ *is an inner product.*

EXERCISE 40. *Verify that* (2) *has the properties of an inner product on* $\mathbb{R}^n$.

EXERCISE 41. *Let* $\mathbb{V}$ *be the space of polynomials with real coefficients. Verify that the following is an inner product on* $\mathbb{V}$:

$$\forall\, p, q \in \mathbb{V} : \quad \langle p, q \rangle = \int_{-1}^{1} p(t) q(t)\, dt. \tag{6}$$

EXERCISE 42. *Let $\mathbb{V} = \mathbb{R}^n$ and fix $w_1, \ldots, w_n > 0$. Verify that the following is an inner product on $\mathbb{V}$:*

$$\forall\, x, y \in \mathbb{V}: \quad \langle x, y\rangle = w_1 x_1 y_1 + \cdots + w_n x_n y_n.$$

The definition of inner product is slightly different for a complex space.

DEFINITION 19. *Let $\mathbb{V}$ be complex vector space. An inner product on V is a map $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{C}$ such that*

(1) $\langle x, x\rangle \geqslant 0$ *for all* $x \in \mathbb{V}$, *and* $\langle x, x\rangle = 0$ *if and only if* $x = 0$.
(2) $\forall\, x, y \in \mathbb{V}: \ \langle x, y\rangle = \overline{\langle y, x\rangle}$.
(3) $\forall\, y \in \mathbb{V}$, *the map* $x \mapsto \langle y, x\rangle$ *is linear.*

REMARK 8. *Property* 2 *is called skew-symmetry. Unlike the inner product on a real vector space, we do not have bilinearity but something called sesquilinearity:*

$$\langle x + y, v\rangle = \langle x, v\rangle + \langle y, v\rangle \quad \text{and} \quad \langle ax, v\rangle = \bar{a}\langle x, v\rangle.$$

EXERCISE 43. *Verify that* (3) *has the properties of an inner product on $\mathbb{C}^n$.*

DEFINITION 20. *A vector space with an inner product is called an inner product space.*

Thus, $\mathbb{R}^n$ and $\mathbb{C}^n$ are examples of inner-product spaces. In fact, we can make the following general statement.

THEOREM 4. *Any finite-dimensional vector space is an inner-product space.*

We now turn to normed spaces. In this case, we do not need to distinguish between real and complex spaces.

DEFINITION 21. *Let $\mathbb{V}$ be a real or complex vector space. An norm on V is a map $\|\cdot\| : \mathbb{V} \to \mathbb{R}$ such that*

(1) $\|x\| \geqslant 0$ *for all* $x \in \mathbb{V}$, *and* $\|x\| = 0$ *if and only if* $x = 0$.
(2) $\forall\, x \in \mathbb{V},\ a \in \mathbb{F}: \ \|ax\| = |a|\,\|x\|$.
(3) $\forall\, x, y \in \mathbb{V},\ \|x + y\| \leqslant \|x\| + \|y\|$.

REMARK 9.

(1) *Property* 1 *is positive definiteness,* 2 *is homogeneity, and* 3 *is triangle inequality.*
(2) *Norm of a vector is a real number, even for a complex vector space.*

EXERCISE 44. *Verify that* (4) *and* (5) *are valid norms on $\mathbb{R}^n$ and $\mathbb{C}^n$.*

DEFINITION 22. *A vector space with a norm is called a normed space.*

Thus, $\mathbb{R}^n$ and $\mathbb{C}^n$ are both inner product and normed spaces. Note that the respective norms (4) and (5) are derived from the respective dot products. In fact, any inner product naturally induces a norm as follows.

EXERCISE 45. *Let $(\mathbb{V}, \langle \cdot, \cdot \rangle)$ be an inner product space. Define*

$$\forall\, x \in \mathbb{V}: \quad \|x\| = \langle x, x\rangle^{\frac{1}{2}}.$$

Verify that $\|\cdot\|$ is a norm. We call $\|\cdot\|$ the norm induced by the inner product $\langle \cdot, \cdot \rangle$.

Combining the above observation with Theorem 4, we have the following.

THEOREM 5. *Any finite-dimensional vector space is a normed space.*

While every inner product space is automatically a normed space, the converse is not true, i.e., not all norms are induced by an inner product, as shown in the following example.

EXERCISE 46. *Let* $\mathbb{V} = \mathbb{R}^2$ *and* $\|x\| = |x_1| + |x_2|$. *Verify that this is a norm. Explain why there does not exist an inner product* $\langle \cdot, \cdot \rangle$ *on* $\mathbb{V}$ *such that* $\|x\| = \langle x, x \rangle^{1/2}$.

We now look at the famous Cauchy-Schwarz inequality relating an inner product to its induced norm.

THEOREM 6. *Let* $(\mathbb{V}, \langle \cdot, \cdot \rangle)$ *be an inner product space with induced norm* $\|\cdot\|$. *Then*

$$\forall\, x, y \in \mathbb{V}: \quad |\langle x, y \rangle| \leqslant \|x\| \cdot \|y\|.$$

A central concept in Euclidean geometry is that of orthogonality. For example, we have the Pythagoras theorem for right-angle triangles. We also have the parallelogram law, which states that the sum of the sides of a parallelogram is twice the sum of the diagonals. These hold in the abstract setting of an inner product space.

In the rest of the discussion, we will assume that $\mathbb{V}$ is an inner product space.

DEFINITION 23. *We say that* $x, y \in \mathbb{V}$ *are orthogonal, denoted* $x \perp y$, *if* $\langle x, y \rangle = 0$. *Furthermore, they are orthonormal if* $\|x\| = \|y\| = 1$.

EXAMPLE 8.

(1) *Vectors* $(1, 1)$ *and* $(1, -1)$ *are orthogonal but not vectors* $(1, 1)$ *and* $(1, 0)$.
(2) *Vectors* $(1, 0)$ *and* $(0, 1)$ *are orthonormal.*
(3) *Consider the inner product* (6). *Then* $p(t) = t$ *and* $q(t) = t^2$ *are orthogonal.*

We can now state the Pythagoras theorem and the parallelogram law.

EXERCISE 47. *Show if that if* $x \perp y$, *then* $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

EXERCISE 48. *Show that* $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$.

REMARK 10. *The proof of the parallelogram law depends crucially on the fact that the norm in question is derived from an inner product. In fact, the parallelogram law need not hold in general normed spaces.*

DEFINITION 24. *A list of vectors* $q_1, \ldots, q_n$ *in* $\mathbb{V}$ *are said to be an orthonormal system if* $\|q_i\| = 1$ *for all* i *and* $q_i \perp q_j$ *for all* $i \neq j$.

EXAMPLE 9.

(1) *The vectors in Exercise 17 form an orthonormal system w.r.t. the dot product.*
(2) *The functions in Exercise 17 are orthonormal wrt the following inner product:*

$$\langle f, g \rangle = \int_0^1 f(t) g(t)\, dt.$$

At this point, we introduce the concept of an orthonormal basis. This is motivated by the following result.

EXERCISE 49. *Show that any orthonormal system of vectors is linearly independent.*

DEFINITION 25. *A basis in which the vectors form an orthonormal system is called an orthonormal basis (ONB).*

EXERCISE 50. *Show that the orthonormal system in Exercise 17 is an ONB of* $\mathbb{R}^n$.

The above ONB is w.r.t. the special inner product on $\mathbb{R}^n$, namely, the dot product. Can we come up with an ONB if we work with a different inner product? What about an abstract inner product?

THEOREM 7. *Any finite-dimensional inner product space has an ONB.*

This can be shown using an inductive process called Gram-Schmidt orthogonalization.

EXERCISE 51. *Let* $\mathbb{V}$ *be the space of polynomials and consider the inner product* (6). *Apply Gram-Schmidt to the polynomials* $p_1(t) = 1$, $p_2(t) = t$, *and* $p_3(t) = t^2$ *to derive an orthonormal system which spans the same space as* p_1, p_2, *and* p_3.

As mentioned earlier, working with an ONB has computational advantages. This is because it is easy to compute the coordinates w.r.t. an ONB.

EXERCISE 52. *Let* $q_1, \ldots, q_n$ *be an ONB of* $\mathbb{V}$. *Prove that, for all* $x \in \mathbb{V}$,

$$x = \sum_{i=1}^{n} \langle x, q_i \rangle q_i \quad \textit{and} \quad \|x\|^2 = \sum_{i=1}^{n} |\langle x, q_i \rangle|^2.$$

We call the latter Parseval's formula; this gives us an easy means of computing the norm of a vector from its coefficients.

Having discussed inner product spaces, we move to linear operators on (and between) such spaces. A fundamental result in this regard is the characterization of the dual space $\mathbb{V}' = \mathcal{L}(\mathbb{V}, \mathbb{F})$ — the space of linear functionals on $\mathbb{V}$.

THEOREM 8. *Let* V *be a finite-dimensional inner product space. Then*

$$\forall\, \ell \in \mathbb{V}',\ \exists\, v \in \mathbb{V}: \quad \ell(x) = \langle v, x \rangle \quad (x \in \mathbb{V}).$$

That is, every linear functional on $\mathbb{V}$ *is given by an inner product.*

EXERCISE 53. *If* $\mathbb{V}$ *is finite-dimensional, then show that* $\mathbb{V}' \cong \mathbb{V}$.

We revisit the concept of a complementary subspace for an inner product space.

DEFINITION 26. *Let* $\mathbb{U}$ *be a subspace of an inner product space* $\mathbb{V}$. *The orthogonal complement of* $\mathbb{U}$ *is defined as*

$$\mathbb{U}^{\perp} = \big\{ x \in \mathbb{V} :\ x \perp u \ \ \forall\, u \in \mathbb{U} \big\}.$$

We record two trivial cases: $\mathbb{V}^{\perp} = \{0\}$ and $\{0\}^{\perp} = \mathbb{V}$.

EXERCISE 54. *Verify that* $\mathbb{U}^{\perp}$ *is a subspace of* $\mathbb{V}$. *Moreover, show that*

$$\mathbb{U} \cap \mathbb{U}^{\perp} = \{0\},\ (\mathbb{U}^{\perp})^{\perp} = \mathbb{U},\ \textit{and}\ \ \mathbb{V} = \mathbb{U} \oplus \mathbb{U}^{\perp}.$$

The last relation is called the orthogonal decomposition of $\mathbb{V}$.

EXERCISE 55. *Let* $q_1, \ldots, q_n$ *be an ONB. For* $1 < k < n$, *let* $\mathbb{U} = \operatorname{span}(q_1, \ldots, q_k)$ *and* $\mathbb{W} = \operatorname{span}(q_{k+1}, \ldots, q_n)$. *Show that* $\mathbb{W} = \mathbb{U}^{\perp}$.

CHAPTER 5

TRANSFORMS ON INNER PRODUCT SPACES

Given the orthogonal decomposition $\mathbb{V} = \mathbb{U} \oplus \mathbb{U}^\perp$, we are naturally led to the concept of an orthogonal projection.

DEFINITION 27. *The orthogonal projection onto a subspace $\mathbb{U}$ is an operator $P_\mathbb{U} : \mathbb{V} \to \mathbb{V}$ defined as*

$$\forall\, x \in \mathbb{V} : \quad P_\mathbb{U}(x) = u \qquad (x = u + v,\ u \in \mathbb{U},\ v \in \mathbb{U}^\perp).$$

REMARK 11.

(1) *It is clear that $P_\mathbb{U} \in \mathcal{L}(\mathbb{V})$.*
(2) *When we say that P is an orthogonal projection, it is understood that $P = P_\mathbb{U}$ for some subspace $\mathbb{U} \subseteq \mathbb{V}$.*

EXERCISE 56. *Let $u \in \mathbb{V}$ and $\mathbb{U} = \operatorname{span}(u)$. Give a formula for $P_\mathbb{U}$.*

Some properties of orthogonal projection are mentioned below.

EXERCISE 57. *Let $P = P_\mathbb{U}$ be an orthogonal projection on $\mathbb{V}$. Show that*

(1) $P^2 = P$.
(2) $\mathcal{R}(P) = \mathbb{U}$ *and* $\mathcal{N}(P) = \mathbb{U}^\perp$.
(3) $P + P_{\mathbb{U}^\perp} = \mathrm{Id}$.
(4) $\|Px\| \leqslant \|x\|$ *for all* $x \in \mathbb{V}$.
(5) *For all* $x \in \mathbb{V}$,
$$P(x) = \operatorname*{argmin}_{u \in \mathbb{U}} \|u - x\|.$$
In other words, $P(x)$ is the (unique) point in $\mathbb{U}$ that is "closest" to x.

EXERCISE 58. *If P is an orthogonal projection, then show that*

$$\forall\, x, y \in \mathbb{V} : \quad \langle Px, y\rangle = \langle x, Py\rangle.$$

We call such operators as self-adjoint. But first, we must define the "adjoint" of an operator; the definition is motivated by the following results.

EXERCISE 59. *For $A \in \mathbb{R}^{m\times n}$, define its transpose $A^\top \in \mathbb{R}^{n\times m}$ to be $(A^\top)_{ij} = A_{ji}$ for all $1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$. Show that*

$$\forall\, x, y \in \mathbb{R}^n : \quad \langle Ax, y\rangle = \langle x, A^\top y\rangle,$$

where $\langle \cdot, \cdot \rangle$ is the standard dot product.

On the other hand, for $A \in \mathbb{C}^{m\times n}$ define its Hermitian transpose $A^\mathrm{H} \in \mathbb{C}^{n\times n}$ to be $(A^\mathrm{H})_{ij} = \overline{A_{ji}}$ for all $1 \leqslant i, j \leqslant n$. Show that

$$\forall\, x, y \in \mathbb{C}^n : \quad \langle Ax, y\rangle = \langle x, A^\mathrm{H} y\rangle,$$

where $\langle \cdot, \cdot \rangle$ is the Hermitian dot product.

In fact, every operator on a finite-dimensional space has an adjoint; this essentially follows from Theorem 8.

THEOREM 9. *Let $\mathbb{V}$ and $\mathbb{W}$ be finite-dimensional and $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$. Prove that there exists an unique operator $T^* \in \mathcal{L}(\mathbb{W}, \mathbb{V})$ such that*

$$\forall\, x \in \mathbb{V},\ y \in \mathbb{W}: \quad \langle Tx, y \rangle = \langle x, T^* y \rangle. \tag{7}$$

We call T^* the adjoint of T. We list some useful properties of the adjoint.

EXERCISE 60. *Let $T, S \in \mathcal{L}(\mathbb{V})$ and $a \in \mathbb{F}$. Show that*

(1) $(T + S)^* = T^* + S^*$ *and* $(aT)^* = \bar{a}T^*$.
(2) $(T^*)^* = T$.
(3) $(TS)^* = S^*T^*$.

We can identify the following four "fundamental subspaces" associated with any $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$:

$$\mathcal{N}(T),\ \mathcal{R}(T^*) \subseteq \mathbb{V} \quad \text{and} \quad \mathcal{N}(T^*),\ \mathcal{R}(T) \subseteq \mathbb{W}.$$

There exists a particular relation between these subspaces.

THEOREM 10. *Let $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$. Then*

(1) $\mathcal{N}(T^*T) = \mathcal{N}(T)$ *and* $\mathcal{N}(TT^*) = \mathcal{N}(T^*)$.
(2) $\mathcal{N}(T)^\perp = \mathcal{R}(T^*)$ *and* $\mathcal{N}(T^*)^\perp = \mathcal{R}(T)$.
(3) $\mathbb{V} = \mathcal{N}(T) \oplus \mathcal{R}(T^*)$ *and* $\mathbb{W} = \mathcal{N}(T^*) \oplus \mathcal{R}(T)$.

As a corollary of the above result, we can conclude the following.

EXERCISE 61. *Prove that for all $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$,* $\operatorname{rank} T^* = \operatorname{rank} T$

EXERCISE 62. *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ be such that the equation $Ax = b$ admits a solution. Show that there is a solution of the form $x = A^\top z$ where $z \in \mathbb{R}^m$.*

EXERCISE 63. *Let $A \in \mathbb{R}^{m \times n}$. Show that the equation $A^\top A = A^\top b$ has a solution for all $b \in \mathbb{R}^m$. Moreover, show that the solution is unique if and only if $\mathcal{N}(A) = \{0\}$.*

We end this part by discussing some important classes of operators that will be studied in greater detail later.

DEFINITION 28. *An operator is self-adjoint if $T^* = T$, i.e., if*

$$\forall\, x, y \in \mathbb{V}: \quad \langle Tx, y \rangle = \langle x, Ty \rangle.$$

In particular, a real self-adjoint matrix ($A^\top = A$) is called symmetric, and a complex self-adjoint matrix ($A^{\mathrm{H}} = A$) is called Hermitian.

REMARK 12. *The inner product is essential in deciding if an operator is self-adjoint. An operator that is self-adjoint w.r.t. one inner product need not be self-adjoint w.r.t. a different inner product.*

EXERCISE 64. *Let $P \in \mathcal{L}(\mathbb{V})$ be such that $P^2 = P$. Prove P is an orthogonal projection if and only if P is self-adjoint.*

EXERCISE 65. *Let $u \in \mathbb{R}^n$. Verify that $A_{ij} = u_i u_j$, $1 \leqslant i, j \leqslant n$, is symmetric. On the other hand, verify that $A_{ij} = v_i \overline{v_j}$, $1 \leqslant i, j \leqslant n$, is Hermitian where $v \in \mathbb{C}^n$.*

EXERCISE 66. *Let $A \in \mathbb{C}^{n \times n}$. Show that if*

$$\forall\, x \in \mathbb{C}^n : \quad \langle x, Ax \rangle_{\mathbb{C}^n} = 0,$$

where $\langle \cdot, \cdot \rangle_{\mathbb{C}^n}$ is the dot product on $\mathbb{C}^n$, then $A = 0$.

On the other hand, give a counterexample to show that if $A \in \mathbb{R}^{2 \times 2}$, then the condition

$$\forall\, x \in \mathbb{R}^2 : \quad \langle x, Ax \rangle_{\mathbb{R}^2} = 0,$$

where $\langle \cdot, \cdot \rangle_{\mathbb{R}^2}$ is the dot product on $\mathbb{R}^2$, does not imply that $A = 0$.

What is special about a self-adjoint operator? We will later see that such operators can always be diagonalized in an ONB. In fact, they form a subclass of a larger set of diagonalizable operators.

DEFINITION 29. *An operator is $T \in \mathcal{L}(\mathbb{V})$ is said to be normal if it commutes with its adjoint: $T^*T = TT^*$.*

Obviously, a self-adjoint operator is normal. In particular, any symmetric or Hermitian matrix is normal. Another important class of normal operators is the unitary operators.

DEFINITION 30. *An operator is $T \in \mathcal{L}(\mathbb{V})$ is said to be unitary if $T^*T = TT^* = \mathrm{Id}$. A real-valued unitary matrix is called an orthogonal matrix.*

In fact, we need just one equality in the above definition if $\mathbb{V}$ is finite-dimensional. This is because of the following observation.

EXERCISE 67. *Let $\mathbb{V}$ be finite-dimensional. Show that $T^*T = \mathrm{Id}$ if and only if $TT^* = \mathrm{Id}$.*

The following is a useful characterization of unitary operators.

EXERCISE 68. *Let $\mathbb{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathbb{V})$. Show that the following are equivalent:*

(1) *T is unitary, i.e., $T^*T = TT^* = \mathrm{Id}$.*
(2) *T is an isomorphism and $T^{-1} = T^*$.*
(3) *$\forall\, x \in \mathbb{V} : \ \|Tx\| = \|x\|$.*
(4) *$\forall\, x, y \in \mathbb{V} : \ \langle Tx, Ty \rangle = \langle x, y \rangle$.*
(5) *T maps any ONB into a (possibly different) ONB.*

EXERCISE 69. *Show that the following operators are unitary.*

(1) *$T \in \mathcal{L}(\mathbb{R}^2)$ given by $T(x_1, x_2) = (x_2, -x_1)$.*
(2) *$A \in \mathbb{R}^{2 \times 2}$ given by*

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

(3) *$A \in \mathbb{C}^{2 \times 2}$ given by*

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{\imath\theta} & -e^{-\imath\theta} \\ e^{\imath\theta} & e^{-\imath\theta} \end{pmatrix}.$$

CHAPTER 6

OPERATOR AND MATRIX NORMS

Similar to vectors, we can define inner products and norms for matrices. After all, the space of matrices (of a fixed shape) is finite-dimensional, and we know that such spaces are automatically an inner product space.

We first look at the concept of operator norm, which can even be defined for operators on infinite-dimensional spaces.

DEFINITION 31. *Let* $(\mathbb{V}, \|\cdot\|_{\mathbb{V}})$ *and* $(\mathbb{W}, \|\cdot\|_{\mathbb{W}})$ *be two normed spaces. The operator (or induced) norm of* $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ *is defined as*

$$\|T\| = \max\Big\{\|Tx\|_{\mathbb{W}} : \ x \in \mathbb{V}, \ \|x\|_{\mathbb{V}} = 1\Big\}. \tag{8}$$

EXERCISE 70. *Verify that* (8) *is a valid norm on* $\mathcal{L}(\mathbb{V}, \mathbb{W})$. *Moreover, show that*

$$\forall\, T, S \in \mathcal{L}(\mathbb{V}) : \quad \|TS\| \leqslant \|T\|\,\|S\|. \tag{9}$$

REMARK 13.

(1) *The sup in* (8) *is finite if* $\mathbb{V}$ *is finite-dimensional but need not be finite if* $\mathbb{V}$ *is infinite-dimensional; if the sup is finite, we call* T *a bounded operator.*
(2) *An important special case is when* T *is a matrix* A *and the norms are Euclidean. In this case, we call* (8) *the spectral norm and is denoted by* $\|A\|_2$. *We will later see that this corresponds to the largest singular value of a matrix.*

Generally, a "matrix norm" is a standard norm on the space of matrices with the additional submultiplicative property (9). We record this for completeness.

DEFINITION 32. *A map* $\|\cdot\| : \mathbb{C}^{n\times n} \to \mathbb{R}$ *is called a matrix norm if it satisfies the following properties.*

(1) $\forall\, A \in \mathbb{C}^{n\times n} : \quad \|A\| \geqslant 0$ and $\|A\| = 0 \iff A = 0$.
(2) $\forall\, A \in \mathbb{C}^{n\times n}, c \in \mathbb{C} : \quad \|cA\| = |c| \cdot \|A\|$.
(3) $\forall\, A, B \in \mathbb{C}^{n\times n} : \quad \|A + B\| \leqslant \|A\| + \|B\|$.
(4) $\forall\, A, B \in \mathbb{C}^{n\times n} : \quad \|AB\| \leqslant \|A\|\,\|B\|$.

We already know from Exercise 70 that the spectral norm is a matrix norm. Another important example is the Frobenius norm.

DEFINITION 33. *The Frobenius norm of* $A \in \mathbb{C}^{m\times n}$ *is defined as*

$$\|A\|_F = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2\right)^{1/2}. \tag{10}$$

EXERCISE 71. *Verify that* (10) *is a matrix norm in the sense of Definition 32 but is not an induced norm in the sense of* (8).

EXERCISE 72. *Let $A \in \mathbb{C}^{m\times n}$ be of rank r. Prove that*

$$\|A\|_2 \leqslant \|A\|_F \leqslant \sqrt{r}\,\|A\|_2.$$

We next look at some other examples of matrix norms.

EXERCISE 73. *For $A \in \mathbb{C}^{m\times n}$, define*

$$\|A\|_1 = \max_{1\leqslant j\leqslant n} \sum_{i=1}^{m} |a_{ij}| \qquad \text{and} \qquad \|A\|_\infty = \max_{1\leqslant i\leqslant m} \sum_{j=1}^{n} |a_{ij}|.$$

Verify that these are norms on $\mathbb{C}^{m\times n}$ and that they are induced by norms on $\mathbb{C}^m$ and $\mathbb{C}^n$.

We know that the series $1 + a + a^2 + \cdots$ converges for any $a \in \mathbb{C}$ if $|a| < 1$. The following is a generalization of this result for matrices. A sequence of matrices $A_1, A_2, \ldots$ is said to converge to A if $\|A_n - A\| \to 0$ as $n \to \infty$ (the choice of matrix norm does not matter).

EXERCISE 74. *Let $A \in \mathbb{C}^{n\times n}$ such that $\|A\| < 1$, where $\|\cdot\|$ is an induced matrix norm. Prove that $I - A$ is invertible, and*

$$\|(I - A)^{-1}\| \leqslant \frac{1}{1 - \|A\|}.$$

Moreover, prove that

$$\lim_{n\to\infty} (I + A + A^2 + \cdots + A^n) = (I - A)^{-1}.$$

The following result tells us that the set of invertible matrices in $\mathbb{C}^{n\times n}$ form an open set.

EXERCISE 75. *Let $A \in \mathbb{C}^{n\times n}$ be invertible and $\|\cdot\|$ be a norm on $\mathbb{C}^{n\times n}$. Show (without using determinants) that there exists $\delta > 0$ such that B is invertible if $\|B - A\| < \delta$.*

We will later need the following result to establish the Cayley-Hamilton theorem.

EXERCISE 76. *Let p be a polynomial with complex coefficients and $\|\cdot\|$ be a norm on $\mathbb{C}^{n\times n}$. Let $A \in \mathbb{C}^{n\times n}$ and (A_n) be a sequence of matrices in $\mathbb{C}^{n\times n}$ such that $\|A_n - A\| \to 0$. Prove that*

$$\lim_{n\to\infty} \|p(A_n) - p(A)\| = 0.$$

CHAPTER 7

TRACE AND DETERMINANT

In this part, we will talk about the trace and determinant of a matrix. These are useful for simplifying calculations involving matrices.

Unlike the previous chapters, we will depart from generic linear transforms and deal exclusively with matrices. In particular, we will use determinants to characterize invertible matrices and find eigenvalues of matrices. Unless otherwise stated, the underlying field $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

DEFINITION 34. *The trace of $A \in \mathbb{F}^{m\times n}$ is $\operatorname{tr}(A) = A_{11} + \cdots + A_{nn}$.*

We can view trace as a linear functional on the space of matrices, i.e., $\operatorname{tr} \in \mathcal{L}(\mathbb{F}^{m\times n}, \mathbb{F})$. In particular, we have the following properties.

EXERCISE 77. *Show that*

(1) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ *for all* $A, B \in \mathbb{F}^{m\times n}$.
(2) $\|A\|_F^2 = \operatorname{tr}(A^{\mathrm{H}}A)$ *for all* $A \in \mathbb{F}^{m\times n}$.
(3) $\|u\|^2 = \operatorname{tr}(uu^\top)$ *for all* $u \in \mathbb{R}^n$.
(4) $\|u\|^2 = \operatorname{tr}(uu^{\mathrm{H}})$ *for all* $u \in \mathbb{C}^n$.
(5) $A \sim B \Rightarrow \operatorname{tr}(A) = \operatorname{tr}(B)$ *for all* $A, B \in \mathbb{F}^{m\times n}$.

EXERCISE 78. *Let A and B be two matrix representations of an operator T, show that $\operatorname{tr}(A) = \operatorname{tr}(B)$. Thus, we can define the trace of T as $\operatorname{tr}(A)$, where $A = [T]$.*

The definition of determinants is more complex. We will do so using permutations. A permutation on $[n] = \{1, 2, \ldots, n\}$ is a reordering of the elements of $[n]$. We can also view them as bijections on $[n]$.

DEFINITION 35. *A permutation on $[n]$ is a bijection $\sigma : [n] \to [n]$. We will use Π_n to denote the set of permutations on $[n]$.*

EXAMPLE 10. $\Pi_1 = \{(1)\}$ *and* $\Pi_2 = \{(1, 2), (2, 1)\}$.

EXERCISE 79. *Compute* $|\Pi_n|$.

A permutation can be decomposed into elementary permutations.

DEFINITION 36. *A permutation $\tau \in \Pi_n$ is called a transposition if there exists $p, q \in [n]$, $p \neq q$, such that*

$$\sigma(i) = \begin{cases} q & i = p, \\ p & i = q, \\ i & i \notin \{p, q\}. \end{cases}$$

We will use $\mathcal{T}_n$ to denote the set of transpositions on $[n]$.

EXAMPLE 11. *$(2,1)$ and $(3,2,1)$ are transpositions but $(2,3,1)$ and $(3,1,2)$ are not transpositions.*

EXERCISE 80. *Compute $|\mathcal{T}_n|$ and compare with $|\Pi_n|$.*

The reason we are interested in transpositions is the following result.

EXERCISE 81. *Show that any permutation is a composition of transpositions, i.e.,*

$$\forall\, \sigma \in \Pi_n,\ \exists\, \tau_1, \dots, \tau_k \in \mathcal{T}_n : \quad \sigma = \tau_1 \circ \cdots \circ \tau_k.$$

EXAMPLE 12. *We can write $(3,1,2)$ as the transposition $(1,2,3) \to (1,3,2)$ followed by the transposition $(1,3,2) \to (3,1,2)$.*

There are different ways to express a permutation using transpositions, e.g., applying the same transposition an even number of times does not make any difference. What is invariant is the parity of the number of transpositions in the decomposition.

EXERCISE 82. *Let $\sigma \in \Pi_n$. Suppose there exists $\tau_1, \dots, \tau_k \in \mathcal{T}_n$ and $\rho_1, \dots, \rho_\ell \in \mathcal{T}_n$ such that*

$$\sigma = \tau_1 \circ \cdots \circ \tau_k = \rho_1 \circ \cdots \circ \rho_\ell.$$

Show that both k and ℓ are even or both are odd. Thus, every $\sigma \in \Pi_n$ has even parity (even permutation) or odd parity (odd permutation).

We encode the above structure using a ± 1 signature.

DEFINITION 37. *The sign of a permutation $\sigma \in \Pi_n$ is defined as*

$$\operatorname{sign}(\sigma) = \Big\{ +1 \qquad \text{if } \sigma \text{ is even}, -1 \qquad \text{if } \sigma \text{ is odd}.$$

EXERCISE 83. *Prove the following results.*

(1) $\operatorname{sign}(\sigma\rho) = \operatorname{sign}(\sigma)\operatorname{sign}(\rho)$ *for all* $\sigma, \rho \in \Pi_n$.
(2) $\Pi_n = \{\sigma\rho : \rho \in \Pi_n\} = \{\rho\sigma : \rho \in \Pi_n\}$ *for all* $\sigma \in \Pi_n$.
(3) $\Pi_n = \{\sigma^{-1} : \sigma \in \Pi_n\}$.

DEFINITION 38. *The determinant of $A = (a_{ij}) \in \mathbb{F}^{n\times n}$ is defined as*

$$\det(A) = \sum_{\sigma \in \Pi_n} \operatorname{sign}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}. \tag{11}$$

We collect some useful properties of determinants.

EXERCISE 84. *Let $A, B \in \mathbb{F}^{m\times n}$. Deduce the following properties from definition* (11).

(1) $\det(A) = \sum_{\sigma\in\Pi_n} \operatorname{sign}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}$.
(2) $\det(I_n) = 1$.
(3) $\forall c \in \mathbb{F} :\ \det(cA) = c^n \det(A)$.
(4) $\det(A^\top) = \det(A)$.
(5) $\det(AB) = \det(A)\det(B)$.
(6) $\det(A) = 0 \iff A$ is not invertible.
(7) $A \sim B \Rightarrow \det(A) = \det(B)$.
(8) A has two identical columns (or rows) $\Rightarrow \det(A) = 0$.
(9) A unitary $\Rightarrow \det(A) = \pm 1$.

CHAPTER 8

DIAGONALIZATION

In this chapter, we will work with a finite-dimensional vector space $\mathbb{V}$ over the field $\mathbb{C}$. We will use $\mathbb{P}_{\mathbb{C}}$ to denote the space of polynomials with complex coefficients.

We wish to understand the conditions for an operator to be "diagonalizable".

DEFINITION 39. *An operator $T \in \mathcal{L}(\mathbb{V})$ is said to be diagonalizable if there exists a basis $v_1, \ldots, v_n$ of $\mathbb{V}$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ such that*

$$\forall\, 1 \leqslant i \leqslant n : \quad Tv_i = \lambda_i v_i.$$

REMARK 14. *The matrix representation $[T]$ with respect to $v_1, \ldots, v_n$ is diagonal:*

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

In other words, a diagonalizable matrix is similar to a diagonal matrix.

Are all operators diagonalizable?

EXERCISE 85. *Let $T \in \mathcal{L}(\mathbb{V})$ be a nonzero operator such that $T^n = 0$ for some $n \geqslant 2$. Show that T cannot be diagonalized. In particular, show that the matrix*

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

cannot be diagonalized as an operator $A \in \mathcal{L}(\mathbb{C}^2)$.

DEFINITION 40. *We say that $\lambda \in \mathbb{C}$ is an eigenvalue of $T \in \mathcal{L}(\mathbb{V})$ if there exists a nonzero $v \in \mathbb{V}$ such that $Tv = \lambda v$. We call v an eigenvector in this case.*

EXAMPLE 13.

(1) *For the zero operator, every nonzero vector is an eigenvector, and 0 is the only eigenvalue*
(2) *For the identity operator, every nonzero vector is an eigenvector, and 1 is the only eigenvalue*
(3) *The eigenvalues of an $(n \times n)$ diagonal matrix are the diagonal elements, and eigenvectors are the columns of the identity matrix I_n.*

EXERCISE 86. *Let T be such that $T^k = 0$ for some $k \geqslant 2$. Find the eigenvalues of T.*

The following relates eigenvalues with determinants. Recall that a square matrix is not invertible if and only if $\det(A) = 0$.

EXERCISE 87. *Let $T \in \mathcal{L}(\mathbb{V})$ and $\lambda \in \mathbb{C}$. Show that λ is an eigenvalue of T if and only if $T - \lambda \mathrm{Id}$ is not injective. In particular, if $A = [T] \in \mathbb{C}^{n\times n}$, then show that $\lambda \in \mathbb{C}$ is an eigenvalue of T if and only if $A - \lambda I_n$ is not invertible, or equivalently,* $\det(A - \lambda I_n) = 0$.

Using the permutation-based formula for determinants, we can deduce the following.

EXERCISE 88. *Let $A \in \mathbb{C}^{n\times n}$. Show that $p(t) = \det(A - tI_n)$ is a polynomial of the form $p(t) = (-1)^n t^n + a_1 t^{n-1} + \cdots + a_n$.*

The polynomial in question is an important object in the study of eigenvalues.

DEFINITION 41. *Let $A = [T] \in \mathbb{C}^{n\times n}$ be a matrix representation of $T \in \mathcal{L}(\mathbb{V})$. We call $p_A(t) = \det(A - tI_n)$ the characteristic polynomial of T.*

EXERCISE 89. *If A and B are the matrix representations of some operator T, then $p_A = p_B$.*

REMARK 15. *An operator on a vector space over the field $\mathbb{R}$ is not guaranteed to have an eigenvalue. Indeed, for the example*

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the characteristic polynomial is $p_A(t) = t^2 + 1$, which has no real roots. However, if we switch to $\mathbb{F} = \mathbb{C}$, then A has two eigenvalues.

We will later need the following result called the Cayley-Hamilton theorem. This requires the concept of a matrix polynomial. For any polynomial $q(t) = a_0 + a_1 t + \cdots + a_m t^m$, $a_i \in \mathbb{C}$, and $A \in \mathbb{C}^{n\times n}$, we define the matrix $q(A) \in \mathbb{C}^{n\times n}$ as follows:

$$q(A) = a_0 I_n + a_1 A + \cdots + a_m A^m.$$

THEOREM 11. *Let $A \in \mathbb{C}^{n\times n}$. Then $p_A(A) = 0$.*

EXERCISE 90. *Let $A \in \mathbb{C}^{n\times n}$ and $q \in \mathbb{P}_{\mathbb{C}}$. Show that if λ is an eigenvalue of A, then $q(\lambda)$ is an eigenvalue of $q(A)$. Conversely, if μ is an eigenvalue of $q(A)$, then show that $\mu = q(\lambda)$ for some eigenvalue λ of A.*

Use the above result to establish the Cayley-Hamilton theorem for diagonalizable matrices.

EXERCISE 91. *Establish Theorem 11 for a diagonalizable matrix.*

We have seen that the degree of the characteristic polynomial of an $(n \times n)$ matrix is n. It follows from the fundamental theorem of algebra that the matrix has precisely n complex eigenvalues, including repetitions.

DEFINITION 42. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. The algebraic multiplicity of T, denoted as $a(\lambda)$, is the number of times λ appears as a root of the characteristic polynomial.*

In other words, we can write the characteristic polynomial as

$$p_{[T]}(t) = (t - \lambda)^{a(\lambda)} q(t), \qquad q(\lambda) \neq 0.$$

That an operator $T \in \mathcal{L}(\mathbb{V})$ can have at most $n = \dim(\mathbb{V})$ distinct eigenvalues can be deduced from the following observation.

EXERCISE 92. *Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T and let $v_1, \dots, v_k$ be the corresponding eigenvectors. Then $v_1, \dots, v_k$ are linearly independent.*

We record the previous fact and an important consequence of it — a sufficient condition for an operator to be diagonalizable.

THEOREM 12. *Let $\mathbb{V}$ be finite-dimensional. Then any $T \in \mathcal{L}(\mathbb{V})$ can have at most $\dim \mathbb{V}$ distinct eigenvalues. In particular, T is diagonalizable if it has $\dim \mathbb{V}$ distinct eigenvalues.*

We remark that the last condition is not necessary for diagonalizability. Indeed, the zero and identity transform have just one distinct eigenvalue but are trivially diagonalizable.

We will now develop a necessary and sufficient condition for an operator to be diagonalizable. We need the concepts of algebraic and geometric multiplicity in this regard.

DEFINITION 43. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. We say that the algebraic multiplicity of λ is k if λ appears k times as a root of the characteristic polynomial. More precisely, we can write the characteristic polynomial as $(t-\lambda)^k q(t)$ for some polynomial q where $q(\lambda) \neq 0$.*

In particular, if $\lambda_1, \dots, \lambda_k \in \mathbb{C}$ are the distinct eigenvalues of an operator, and $a_1, \dots, a_k$ are the corresponding algebraic multiplicities, then the characteristic polynomial must be of the form

$$p(t) = (-1)^n (t-\lambda_1)^{a_1} \cdots (t-\lambda_k)^{a_k}.$$

DEFINITION 44. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. We call $(T - \lambda \operatorname{Id})$ the eigenspace of λ and it geometric multiplicity is defined as $g(\lambda) = \operatorname{nullity}(T - \lambda \operatorname{Id})$.*

EXERCISE 93. *Let $\lambda_1, \dots, \lambda_k \in \mathbb{C}$ be the distinct eigenvalue of $T \in \mathcal{L}(\mathbb{V})$. Let $u_1, \dots, u_k$ be such that $u_i \in \mathcal{N}(T - \lambda_i \operatorname{Id})$ for $i = 1, \dots, k$. Prove that*

$$u_1 + \cdots + u_k = 0 \Rightarrow u_1 = \cdots = u_k = 0.$$

A natural question is whether there is a relation between algebraic and geometric multiplicity of an eigenvalue.

EXERCISE 94. *Show that the algebraic multiplicity of an eigenvalue is at least as large as its geometric multiplicity. Moreover, using the example*

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{12}$$

show that the algebraic multiplicity can be larger than the geometric multiplicity.

We say that an operator is "defective" if the algebraic multiplicity of one of its eigenvalues is strictly larger than its geometric multiplicity. The following result tells us that these are precisely the operators that cannot be diagonalized.

THEOREM 13. *Let $\lambda_1, \dots, \lambda_k \in \mathbb{C}$ be the distinct eigenvalue of $T \in \mathcal{L}(\mathbb{V})$. Prove that the following are equivalent.*

(1) $g(\lambda_1) + \cdots + g(\lambda_k) = \dim \mathbb{V}$.

(2) $\forall\, 1 \leqslant i \leqslant k:\ a(\lambda_i) = g(\lambda_i)$.
(3) *T is diagonalizable.*

As a corollary of Theorem 13, we get the sufficient condition in Theorem 12.

We conclude this section with a useful result called the Schur decomposition. We have seen that not all matrices are diagonalizable. However, every matrix is similar to an upper triangular matrix.

THEOREM 14. *For any $A \in \mathbb{C}^{n\times n}$, there exists an unitary matrix $U \in \mathbb{C}^{n\times n}$ and an upper triangular matrix $\Lambda \in \mathbb{C}^{n\times n}$ such that $A = U\Lambda U^{\mathrm{H}}$.*

This is called the Schur decomposition. It relies on the existence of eigenvalues of a complex matrix. By upper triangular, we mean $\Lambda_{ij} = 0$ for $i > j$.

EXERCISE 95. *Show that the diagonal elements of Λ are the eigenvalues of A.*

We will establish the Schur decomposition using induction on n. We will use the following base case.

EXERCISE 96. *Establish the Schur decomposition for any $A \in \mathbb{C}^{2\times 2}$.*

We will next deduce the Cayley-Hamilton theorem using a limiting argument and the Schur decomposition.

LEMMA 2. *Diagonalizable matrices are dense in $\mathbb{C}^{n\times n}$.*

By dense, we mean that for any $A \in \mathbb{C}^{n\times n}$, there exists a sequence of diagonalizable matrices A_n such that $\|A_n - A\| \to 0$ as $n \to \infty$, where $\|\cdot\|$ is some matrix norm.

Using Lemma 2 and Exercise 76, we can establish the Cayley-Hamilton theorem for general matrices.

CHAPTER 9

EIGEN DECOMPOSITION

Recall the concept of a self-adjoint operator on an inner product space. This includes, in particular, symmetric and Hermitian matrices. They are, in turn, a particular case of normal operators, which include skew-adjoint and unitary operators. We will show that these operators can be diagonalized in an orthonormal (or unitary) basis. Notably, we will develop this theory without using determinants, characteristic equations, or the fundamental theorem of algebra.

THEOREM 15. *Let $\mathbb{V}$ be a complex, finite-dimensional inner product space, and let $T \in \mathcal{L}(\mathbb{V})$ be self-adjoint. Then there exists an orthonormal basis of eigenvectors of T.*

REMARK 16.

(1) *We can deduce that the eigenvalues of T are real.*
(2) *The above result is called the spectral decomposition of T.*
(3) *That the eigenvalues are real is a bonus and is not strictly required for T to be diagonalizable.*
(4) *Specializing the above result to a symmetric matrix A, we can conclude the existence of a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and an orthogonal matrix Q such that*

$$A = Q\Lambda Q^\top. \tag{13}$$

On the other hand, if $A \in \mathbb{C}^{n\times n}$ is Hermitian, then we can write

$$A = U\Lambda U^{\mathrm{H}}, \tag{14}$$

where U is unitary and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \in \mathbb{R}$. We refer to (13) *and* (14) *as the eigendecomposition of A.*

Theorem 15 can be deduced via induction on the dimension of the space, along with the following result. That the eigenvalues are real is a part of the proof.

LEMMA 3. *Every self-adjoint operator has an eigenvector with a real eigenvalue.*

The induction process is based on the concept of an invariant subspace.

DEFINITION 45. *A subspace $\mathbb{U} \subset \mathbb{V}$ is an invariant subspace of $T \in \mathcal{L}(\mathbb{V})$ if $Tu \in \mathbb{U}$ for all $u \in \mathbb{U}$.*

EXERCISE 97. *Let $T \in \mathcal{L}(\mathbb{V})$ be self-adjoint and $\mathbb{U}$ be an invariant subspace of T. Define $T_{\mathbb{U}} \in \mathcal{L}(\mathbb{U})$, the restriction of T on U, as follows:*

$$\forall\, u \in \mathbb{U}: \quad T_{\mathbb{U}} u = Tu.$$

Show that $T_{\mathbb{U}} \in \mathcal{L}(\mathbb{U})$ is self-adjoint.

The archetype example of invariant subspaces is an eigenspace and its orthogonal complement.

EXERCISE 98. *Let q be an eigenvector of T. Show that T is invariant on* $\operatorname{span}\{q\}$ *and* $\operatorname{span}\{q\}^{\perp}$.

We can extend the spectral theorem to the larger class of normal operators. However, the eigenvalues are no longer guaranteed to be real.

THEOREM 16. *Let $\mathbb{V}$ be a complex, finite-dimensional inner product space, and let $T \in \mathcal{L}(\mathbb{V})$ be a normal operator. Then there exists an orthonormal basis of eigenvectors of T.*

Theorem 16 can be deduced from the following observation, which is an important result in itself.

EXERCISE 99. *Let $\mathbb{V}$ be a finite-dimensional inner product space. Suppose $T, S \in \mathcal{L}(\mathbb{V})$ be self-adjoint and $TS = ST$. Then T and S can be diagonalized in a common orthonormal basis.*

We next look at a variational characterization of eigenvalues called the min-max theorem. It allows us to define eigenvalues without the need to associate eigenvectors with them.

THEOREM 17. *Let $\mathbb{V}$ be a finite-dimensional inner product space. Let T be self-adjoint and let $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$ be its eigenvalues. Then, for all $1 \leqslant k \leqslant n$,*

$$\lambda_k = \max_{\dim \mathbb{U} = k} \min \Big\{ \langle x, Tx \rangle : \ x \in \mathbb{U}, \ \|x\| = 1 \Big\},$$

where $\mathbb{U}$ is a subspace of $\mathbb{V}$. In particular,

$$\lambda_1 = \max_{x \in \mathbb{V}, \, \|x\| = 1} \langle x, Tx \rangle \qquad \text{and} \qquad \lambda_n = \min_{x \in \mathbb{V}, \, \|x\| = 1} \langle x, Tx \rangle.$$

Using Theorem 17, establish the perturbation result in Problem 17.

We conclude this part by discussing a particular class of self-adjoint operators that come up in various applications.

DEFINITION 46. *A self-adjoint operator is said to be positive semidefinite (resp. positive definite) if the eigenvalues of T are nonnegative (resp. positive).*

EXAMPLE 14. *Following are some examples of positive semidefinite (PSD) and positive definite (PD) operators.*

(1) *The zero operator is PSD, while the identity operator is PD.*
(2) *For all $q \in \mathbb{V}$, the operator $T(x) = \langle x, q \rangle \, q$ is PSD.*
(3) *Hessian matrix of a twice continuously differentiable convex function.*
(4) *Gram and covariance matrices associated with a system of vectors.*

The following is a useful characterization of PSD and PD operators.

EXERCISE 100. *Let $T \in \mathcal{L}(\mathbb{V})$ be self-adjoint. Show that*

$$T \text{ is PSD} \iff \forall\, x \in \mathbb{V} : \ \langle x, Tx \rangle \geqslant 0,$$

and

$$T \text{ is PD} \iff \forall\, x \in \mathbb{V} \setminus \{0\} : \ \langle x, Tx \rangle > 0.$$

CHAPTER 10

SINGULAR VALUE DECOMPOSITION

Let $A \in \mathbb{C}^{m\times n}$. The question of diagonalization does not arise if $m \neq n$ (rectangular matrix). In fact, we have even seen examples of square matrices which cannot be diagonalized. The next best is the singular value decomposition (SVD) of A.

THEOREM 18. *Let $A \in \mathbb{C}^{m\times n}$ and $r = \operatorname{rank} A \geqslant 1$. Then there exists orthonormal vectors $v_1, \dots, v_r \in \mathbb{C}^n$, orthonormal vectors $u_1, \dots, u_r \in \mathbb{C}^m$, and $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$, such that*

$$A = \sigma_1 u_1 v_1^\top + \cdots + \sigma_r u_r v_r^\top. \tag{15}$$

REMARK 17.

(1) *By extending the orthonormal vectors into orthonormal bases, we can write*

$$A = U\Sigma V^\top, \tag{16}$$

where $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{m\times n}$ is defined as

$$\Sigma_{ij} = \begin{cases} \sigma_i & 1 \leqslant i = j \leqslant r, \\ 0 & \text{otherwise.} \end{cases}$$

We call (15) *the partial SVD of A and* (16) *the full SVD.*

(2) *We call $\sigma_1, \dots, \sigma_r$ the singular values, $u_1, \dots, u_r$ the left singular vectors, and $v_1, \dots, v_r$ the right singular vectors.*

(3) *The SVD and eigendecomposition of A need not coincide even if A is Hermitian. Indeed, while eigenvalues can take negative values, singular values are nonnegative by construction. However, they are the same if A is PSD.*

EXERCISE 101. *Prove that for $A \in \mathbb{C}^{m\times n}$,*

$$\sigma_1(A) = \max\left\{\|Ax\|_{\mathbb{C}^m} : \|x\|_{\mathbb{C}^n} = 1\right\},$$

where $\|\cdot\|_{\mathbb{C}^m}$ and $\|\cdot\|_{\mathbb{C}^n}$ are the Euclidean norms on $\mathbb{C}^m$ and $\mathbb{C}^n$.

In other words, $\sigma_1(A) = \|A\|_2$, the spectral norm induced by the standard Euclidean norms on $\mathbb{C}^n$ and $\mathbb{C}^m$.

Consider the full SVD (16). Let $v_1, \dots, v_n \in \mathbb{C}^n$ be the columns of V and $u_1, \dots, u_m \in \mathbb{C}^m$ be the columns of U. The following result relates these to the four fundamental subspaces of A.

EXERCISE 102. *Show that $\mathcal{R}(A) = \operatorname{span}(u_1, \dots, u_r)$ and $\mathcal{N}(A) = \operatorname{span}(v_{r+1}, \dots, v_n)$.*

Similar to Theorem 17, we have the following min-max characterization.

EXERCISE 103. *Show that, for all* $1 \leqslant k \leqslant r$,

$$\sigma_k = \min_{\dim \mathbb{U}=k} \max \left\{ \|Ax\|_2 : \ x \in \mathbb{U}, \ \|x\|_2 = 1 \right\},$$

where $\mathbb{U}$ *is a subspace of* $\mathbb{V}$.

We have the following perturbation result as an application of the min-max characterization.

EXERCISE 104. *Let* $A \in \mathbb{C}^{n\times n}$. *Let* $\operatorname{rank}(A) = r$ *and* $\sigma_1 \geqslant \cdots \geqslant \sigma_r$ *be the singular values of* A. *If* $\|E\|_2 < \sigma_r$, *then prove that*

$$\operatorname{rank}(A + E) \geqslant r.$$

This means that a "small" perturbation cannot reduce the rank.

We can use the SVD to define the two important matrix norms—the spectral norm (largest singular value) and the nuclear norm (sum of the singular values).

EXERCISE 105. *Let* (15) *be the SVD of* $A \in \mathbb{C}^{m\times n}$. *Define* $\|A\|_2 = \sigma_1$ *and* $\|A\|_* = \sigma_1 + \cdots + \sigma_r$. *Verify that these are matrix norms on* $\mathbb{C}^{m\times n}$. *Moreover, prove that these are dual of each other in the sense that*

$$\|A\|_* = \max_{\|X\|_2 \leqslant 1} \operatorname{tr}(X^\top A),$$

and

$$\|A\|_2 = \max_{\|X\|_* \leqslant 1} \operatorname{tr}(X^\top A),$$

where the variable $X \in \mathbb{C}^{m\times n}$.

We can use the SVD for low-rank approximation. In particular, we have the following result which states that the optimal rank-k approximation of a matrix is provided by a k-term truncation of (15).

EXERCISE 106. *Let* (15) *be the SVD of* $A \in \mathbb{C}^{m\times n}$ *and let* $k \leqslant r$. *Define*

$$A_k = \sigma_1 u_1 v_1^\top + \cdots + \sigma_k u_k v_k^\top.$$

Prove that

$$\|A_k - A\|_* = \max_{\operatorname{rank}(X) \leqslant k} \|X - A\|_*,$$

where $\|\cdot\|_*$ *is the nuclear norm.*

CHAPTER 11

POSITIVE MATRICES

In this chapter, we will work with real square matrices. However, for eigenvalue computation, we will consider them as elements of $\mathbb{C}^{n\times n}$. We say that A is positive ($A > 0$) if each $A_{ij} > 0$ and that A is nonnegative ($A \geqslant 0$) if each $A_{ij} \geqslant 0$. Similarly, we will use the symbols $x > 0$ and $x \geqslant 0$ for vectors. We will use $\sigma(A)$ for the spectrum of A, the set of eigenvalues of A. The spectral radius of A is defined as

$$\rho(A) = \max\ \big\{|\lambda| :\ \lambda \in \sigma(A)\big\}.$$

To motivate the Perron-Frobenius theorem, it would be instructive to look at a 2×2 matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{17}$$

where a, b, c, and d are positive. The eigenvalues of A are

$$\frac{1}{2}(a + d \pm \Delta), \qquad \Delta = \big((a-d)^2 + 4bc\big)^{1/2}.$$

Since $\Delta > 0$, we can conclude that

- $r = \rho(A) = (a + d + \Delta)/2$ is positive and $r \in \sigma(A)$.
- r is the only eigenvalue such that $|\lambda| = r$.
- The algebraic multiplicity of r is one.
- The eigenvector corresponding to r is $p = (a - d + \Delta, 2c)$, a positive vector.
- The other eigenvector $(2b, d - a - \Delta)$ has a negative component.

In fact, any positive matrix has a positive eigenvalue (Perron root) and a corresponding positive eigenvector (Perron vector). Moreover, any nonnegative eigenvector must be a multiple of the Perron vector.

THEOREM 19. *Let $A > 0$ and $r = \rho(A)$. Then*

(1) *$r > 0$ and r is an eigenvalue of A.*
(2) *r is the only eigenvalue on the spectral circle $\{\lambda \in \mathbb{C} : |\lambda| = r\}$.*
(3) *There exists $p > 0$ such that $Ap = rp$.*
(4) *The algebraic (hence the geometric) multiplicity of r is one.*
(5) *If x is an eigenvector of A and $x \geqslant 0$, then $x = cp$, $c > 0$.*

REMARK 18. *We call r the Perron root of A. The Perron vector is the unique vector $p > 0$ such as $Ap = rp$ and $\|p\|_1 = 1$.*

Among other things, we will need the following results to establish Theorem 19.

EXERCISE 107. *Let $\|\cdot\|$ be an induced matrix norm on $\mathbb{C}^{n\times n}$. Prove that*

$$\forall A \in \mathbb{C}^{n\times n}: \quad \rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}.$$

EXERCISE 108. *Let $A \in \mathbb{C}^{n\times n}$ be such that $\rho(A) < 1$. Then, for any matrix norm $\|\cdot\|$,*

$$\lim_{k\to\infty} \|A^k\| = 0.$$

PROPOSITION 6. *Let $A > 0$ and $\rho(A) = 1$. Suppose $q > 0$ be such that $q \leqslant Aq$. Show that $q = Aq$.*

What can be said if A is not positive but nonnegative, i.e., has some zeros? Clearly, r cannot be guaranteed to be positive since this class includes the 0 matrix. A more interesting example is

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{18}$$

In this case, $\sigma(A) = \{0\}$, $\rho(A) = 0$, the algebraic multiplicity of 0 is two, and its eigenvectors are multiples of $(1, 0)$. Thus, (18) violates Theorem 19 on multiple counts. However, we can guarantee the following.

EXERCISE 109. *Let $A \geqslant 0$ and $r = \rho(A)$. Prove that*

(1) $r \in \sigma(A)$.
(2) *There exists $x \geqslant 0$, $x \neq 0$, such that $Ax = rx$.*

This is far as Theorem 19 can be generalized without additional assumptions. Frobenius noticed that depending on the "position of the zeros", one can recover almost all the properties in Theorem 19. For example, consider the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{19}$$

In this case, $\sigma(A) = \{-1, 1\}$ and $\rho(A) = 1$. Moreover, the positve vector $p = (1, 1)$ corresponds to the eigenvalue $\lambda = 1$. Thus, (19) satisfies properties 1, 3, and 4 in Theorem 19.

DEFINITION 47. *A matrix A is said to be reducible if there exists a permutation matrix P and matrices X, Y and Z such that*

$$P^\top A P = \begin{pmatrix} X & Y \\ 0 & Z \end{pmatrix} \tag{20}$$

Otherwise, A is said to be irreducible.

The pre-multiplication with $P^\top$ and post-multiplication with P has the effect of reordering the rows and columns of A. Thus, (18) is reducible, while (19) is irreducible.

EXERCISE 110. *Prove that the following $(n \times n)$ tridiagonal matrix is irreducible:*

$$\begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \ddots & \vdots \\ 0 & 1 & 1 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 1 \end{pmatrix}.$$

We can characterize the irreducibility of $A \in \mathbb{R}^{n\times n}$ based on its graphical representation. Let $\mathcal{G}(A)$ be a directed graph with nodes $\{1, 2, \ldots, n\}$ that has an edge $i \to j$ if and only if $A_{ij} > 0$. A path from node i to node j is a sequence of nodes $i = i_0, i_1, \ldots, i_n = j$ such that $i_k \to i_{k+1}$ for all $k = 0, \ldots, n-1$. We say that $\mathcal{G}(A)$ is strongly connected if, for every pair of nodes (i, j), there exists a path from i to j.

EXERCISE 111. *Prove that A is irreducible if and only if $\mathcal{G}(A)$ is strongly connected.*

THEOREM 20. *Let $A \geqslant 0$ be irreducible and let $r = \rho(A)$. Then*

(1) *$r > 0$ and $r \in \sigma(A)$.*
(2) *The algebraic multiplicity of r is one.*
(3) *There exists $p > 0$ such that $Ap = rp$.*
(4) *The only nonnegative eigenvectors are p and its positive multiples.*

REMARK 19. *The only property that irreducibility cannot salvage is property 2 in Theorem 19. Indeed, we have two different eigenvalues of* (19) *on the spectral circle $|\lambda| = 1$.*

EXERCISE 112. *Verify that the tridiagonal matrix*

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

satisfies the properties in Theorem 20.

We can deduce Theorem 20 from Theorem 19 using the following observations.

EXERCISE 113. *Let $A, B \in \mathbb{R}^{n\times n}$ be such that $A \geqslant B$. Show that $\rho(A) \geqslant \rho(B)$.*

EXERCISE 114. *Let $A > 0$ and define $A_k = A + k^{-1}E$ for all $k \geqslant 1$, where E is the all-ones matrix. Prove that*

$$\lim_{k\to\infty} \rho(A_k) = \rho(A).$$

EXERCISE 115. *Let $A \geqslant 0$ be irreducible. Show that, for all $1 \leqslant i, j \leqslant n$, there exists $1 \leqslant k \leqslant n-1$ such that $(A^k)_{ij} > 0$. In particular, show that $(I + A)^{n-1} > 0$.*

EXERCISE 116. *Let $A \in \mathbb{C}^{n\times n}$ be p be polynomial with complex coefficients. Show that*

$$\sigma(p(A)) = \{p(\lambda) : \lambda \in \sigma(A)\}.$$

EXERCISE 117. *Let $A \in \mathbb{C}^{n\times n}$ be such that $\rho(A) \in \sigma(A)$. Show that, for all $k \geqslant 1$,*

$$\rho\big((I + A)^k\big) = (1 + \rho(A))^k.$$

CHAPTER 12

APPLICATIONS

1. Linear regression

Linear regression is a powerful and versatile technique widely employed in data analysis, machine learning, and scientific research. This uses a linear (affine) model to capture the relationship between dependent and independent variables. We will consider a simple yet profound application of linear regression, namely, fitting a polynomial using linear regression. In many real-world scenarios, the relationship between variables is not linear. Polynomial regression can be used to capture these nonlinear relationships by fitting a polynomial equation to the data.

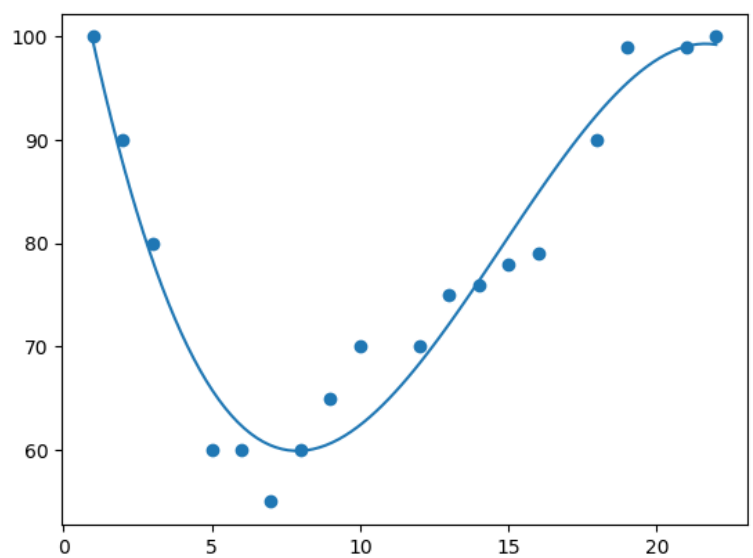

FIGURE 1. Fitting a cubic polynomial.

A polynomial of degree n can be expressed as

$$p(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \ldots + \beta_n x^n,$$

where y is the dependent variable, x is the independent variable, and $\beta_0, \ldots, \beta_n$ are the coefficients. Suppose we are given data points $(x_1, y_1), \ldots, (x_K, y_K)$. The fitting problem is to find a polynomial that best explains the data. In other words, we wish to find $\beta_0, \ldots, \beta_n$ such that $p(x_k) \approx y_k$ for all $k = 1, \ldots, K$. We use "$\approx$" since there might not exist a p that fits each point exactly (think of fitting three non-collinear

points with a line). In least-squares regression, we wish to find $\beta_0, \ldots, \beta_n$ that minimizes

$$\ell(\beta_0, \ldots, \beta_n) = \sum_{k=1}^{K} (y_k - p(x_k))^2.$$

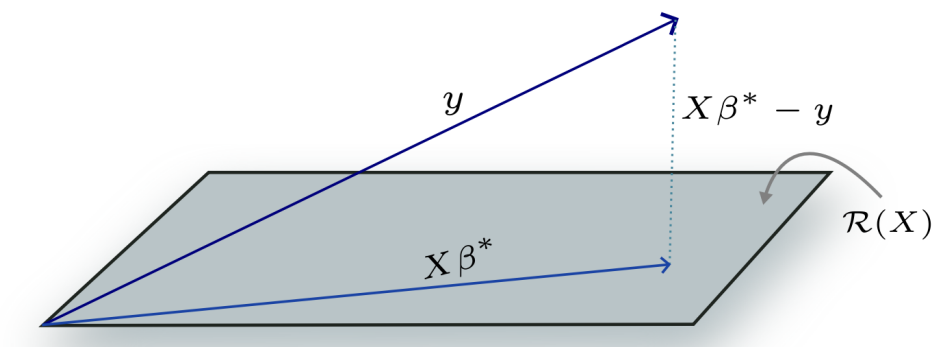

FIGURE 2. The normal equation can be written as $X^\top(X\beta^* - y) = 0$ which requires the optimal residual vector $X\beta^* - y$ to be perpendicular to the range space of X. The latter property is essentially a consequence of the Pythagoras theorem.

Despite the nonlinear nature of p, we can find the optimal set of coefficients that minimizes (1) using linear algebra. More precisely, define the coefficient vector $\beta = (\beta_0, \ldots, \beta_n)$, the measurement vector $y = (y_1, \ldots, y_K)$, and the data matrix

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & \ldots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_K & x_K^2 & \ldots & x_K^n \end{pmatrix}.$$

We can then express the regression problem as

$$\min_{\beta \in \mathbb{R}^{n+1}} \ell(\beta) = \|y - X\beta\|^2, \tag{21}$$

where $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^K$. This is an instance of an unconstrained optimization problem, where the loss function is quadratic. We wish to find β^* such that $\ell(\beta^*) \leqslant \ell(\beta)$ for all β; we call β^* a minimizer of (21). The following result connects this problem with the solution of linear systems of equations.

EXERCISE 118. *Prove that β^* is a minimizer of* (21) *if and only if $X^\top X\beta^* = X^\top y$.*

That is, we can find the minimizer by solving $X^\top X\beta = X^\top y$. This is called the normal equations and has a nice geometric interpretation (see fig. 2)

How do we know that these equations have a solution in the first place? Fortunately, it does. We can use the decomposition range-nullspace decomposition of a linear transform to establish the following result.

EXERCISE 119. *Prove that, for any X and y, there exists β such that $X^\top X\beta = X^\top y$.*

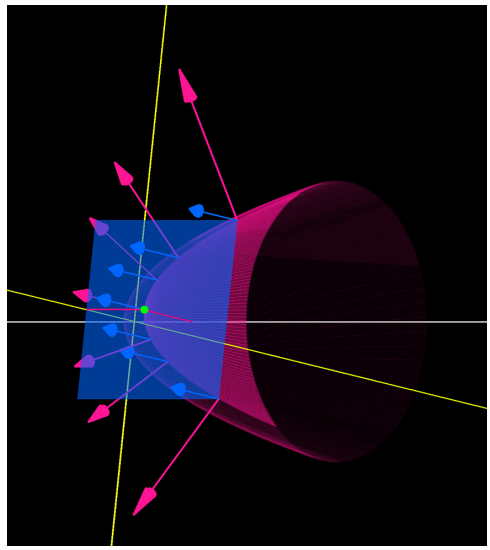

FIGURE 3. Optimizing a quadratic function over an affine space.

In other words, we can always solve the regression problem (21) irrespective of the nature of the data points.

2. Linearly constrained optimization

We often encounter the problem of minimizing a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ over an affine space $Ax = b$ ($A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$):

$$\min_{\beta \in \mathbb{R}^n} f(\beta) \quad \text{subject to } A\beta = b. \tag{22}$$

This is an example of a constrained optimization problem. Unlike linear regression, the variables in this case are restricted and cannot take on all possible values. However, note that the regression problem is a special case of (22): on setting $A = 0$ and $b = 0$ (all $\beta \in \mathbb{R}^n$ is admissible) and $f(\beta) = \|X\beta - y\|^2$ in (22), we obtain (21).

We can characterize the solution of (22) using linear algebra.

EXERCISE 120. *Let $\beta^* \in \mathbb{R}^n$ be a solution of* (22). *Prove that there exists $\lambda \in \mathbb{R}^m$ s.t.*

$$\nabla f(\beta^*) = A^\top \lambda. \tag{23}$$

Consider the particular case

$$f(\beta) = \frac{1}{2}\beta^\top Q \beta, \tag{24}$$

where Q is a symmetric matrix. Then (23) reduces to $Q\beta^* = A^\top \lambda$. Also, since $A\beta^* = b$, we obtain the system of equations

$$\begin{pmatrix} A & 0 \\ Q & -A^\top \end{pmatrix} \begin{pmatrix} \beta^* \\ \lambda \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}. \tag{25}$$

This constitutes $m + n$ equations in $m + n$ variables. Assuming we can solve this system of equations, can we assert that β^* is the desired minimizer? Unfortunately, (23) is just a necessary condition and the answer is generally no. However, if Q is positive semidefinite (i.e., f is convex), then β^* is indeed the desired minimizer.

EXERCISE 121. *Let β^* and λ be a solution of* (25) *and f be given by* (24) *where Q is PSD. Prove that β^* is the solution of* (22).

3. Subspace approximation

We consider the problem of approximating a set of points $x_1, \ldots, x_K$ in $\mathbb{R}^d$ by a subspace $\mathbb{V} \subset \mathbb{R}^d$. This comes up in dimensionality reduction techniques (where $\dim \mathbb{V} \ll d$) such as principal component analysis. We assume the points are centered around the origin.

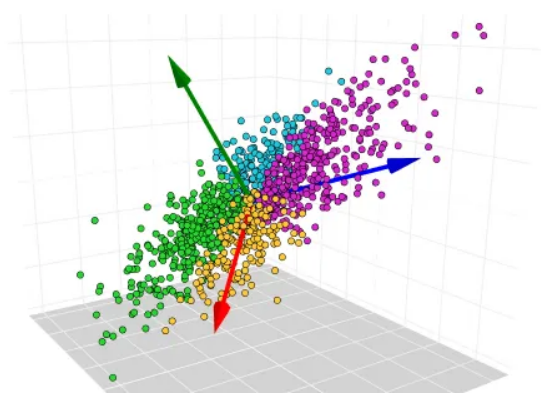

FIGURE 4. Approximating a point cloud in $\mathbb{R}^3$ by a plane. This is done by computing the eigenbasis of the covariance matrix.

The mathematical model for this approximation problem can be viewed as a generalization of linear regression to higher dimensions; however, there is a subtle difference in the way the loss function is defined. More precisely, let $P_{\mathbb{V}} \in \mathcal{L}(\mathbb{R}^d)$ be the orthogonal projector onto the subspace $\mathbb{V}$. The "best" approximation of each x_k would be $P_{\mathbb{V}}(x_k)$ and the distortion incurred is the projection error $\|x_k - P_{\mathbb{V}} x_k\|$, where $\|\cdot\|$ ie the standard Eulcidean norm. On summing the errors from all the points, we obtain the loss function

$$\ell(\mathbb{V}) = \frac{1}{K} \sum_{k=1}^{K} \|x_k - P_{\mathbb{V}} x_k\|^2, \tag{26}$$

where the variable is the subspace $\mathbb{V}$. Thus, finding the optimal subspace amounts to minimizing (26) w.r.t $\mathbb{V}$.

We can solve the above problem by representing $\mathbb{V}$ in an orthonormal basis and expressing (26) in terms of this basis vectors. More precisely, let $\dim \mathbb{V} = p$ and $\mathbb{V} = \operatorname{span}(v_1, \ldots, v_p)$, where $v_1, \ldots, v_p$ are is an orthonormal basis of $\mathbb{V}$.

EXERCISE 122. *For any $x \in \mathbb{R}^d$, show that*

$$\|x - P_{\mathbb{V}} x\|^2 = \|x\|^2 - \sum_{j=1}^{p} (x^\top v_j)^2. \tag{27}$$

In particular, define the covariance matrix

$$C = \frac{1}{K} \sum_{k=1}^{K} x_k x_k^\top.$$

Explain using (27) *why minimizing* (26) *is the same as maximizing*

$$\sum_{j=1}^{p} v_j^\top C v_j \tag{28}$$

where the variable is the orthonormal basis $v_1, \ldots, v_p$.

In other words, we have reduced the original subspace approximation problem to an eigenvalue problem. This has a standard solution. Notice that C is a PSD matrix with nonnegative eigenvalues.

EXERCISE 123. *Prove that* (28) *is maximized if we take $v_1, \ldots, v_p$ to be the top eigenvectors of C, i.e., the eigenvectors corresponding to the largest eigenvalues of C.*

A popular technique for reducing the number of features in a dataset without losing important information is the principal component analysis (PCA). This is modeled on the approximation problem discussed above. Based on the above analysis, we can summarize the key steps in PCA:

(1) Center the data to have zero mean (e.g. by subtracting the mean).
(2) Compute the covariance matrix C of the centered data.
(3) Find the top-p eigenvectors of C (e.g. using the power method).
(4) Project the data onto the subspace spanned by these eigenvectors.

4. Low-rank approximation

5. Closest orthogonal transform

6. Markov chain

7. Convolutions and DFT

CHAPTER 13

PROBLEMS

PROBLEM 1. *Let $\mathbb{K}$ denote the set of real-valued functions on $[0,1]$. With the natural vector space operations on $\mathbb{K}$, explain why $\mathbb{K}$ is infinite-dimensional. On the other hand, what can we say about the dimension of the vector space of functions $f : \{0,1\} \to \mathbb{R}$?*

PROBLEM 2. *Consider the space $\mathbb{P}$ of polynomials of degree at most n. Verify that $p_0(t) = 1, p_1(t) = t, \ldots, p_n(t) = t^n$ form a basis for $\mathbb{P}$.*

Consider the subspace

$$\mathbb{Q} = \Big\{ p \in \mathbb{P} : \int_0^1 p(t)\, dt = 0 \Big\}.$$

Find the dimension of $\mathbb{Q}$.

PROBLEM 3. *Let $\mathbb{V}$ be a vector space and $v_1, \ldots, v_n \in \mathbb{V}$ be linearly independent. Show that for any $v \in \mathbb{V}$, the dimension of $\operatorname{span}(v_1 + v, \ldots, v_n + v)$ is at least $n - 1$.*

PROBLEM 4. *Let $\mathbb{U}_1$ and $\mathbb{U}_2$ be subspaces of a finite-dimensional inner product space $\mathbb{V}$. Prove that $(\mathbb{U}_1^\perp)^\perp = \mathbb{U}_1$ and $(\mathbb{U}_1 \cap \mathbb{U}_2)^\perp = \mathbb{U}_1^\perp + \mathbb{U}_2^\perp$.*

PROBLEM 5. *Let $\mathbb{U}_1$ and $\mathbb{U}_2$ be subspaces of a finite-dimensional space. Show that*

$$\dim\, \mathbb{U}_1 + \mathbb{U}_2 = \dim \mathbb{U}_1 + \dim \mathbb{U}_2 - \dim\, \mathbb{U}_1 \cap \mathbb{U}_2.$$

In particular, if $\mathbb{U}_1$ and $\mathbb{U}_2$ form a direct sum, then show that

$$\dim\, \mathbb{U}_1 \oplus \mathbb{U}_2 = \dim \mathbb{U}_1 + \dim \mathbb{U}_2.$$

PROBLEM 6. *Let $\mathbb{U}_1, \mathbb{U}_2, \mathbb{U}_3$ be subspaces of an n-dimensional vector space. Prove that*

$$\dim\, (\mathbb{U}_1 \cap \mathbb{U}_2 \cap \mathbb{U}_3) \geqslant \dim\, \mathbb{U}_1 + \dim\, \mathbb{U}_2 + \dim\, \mathbb{U}_3 - 2n.$$

PROBLEM 7. *Let $\mathbb{V}$ be a finite-dimensional real vector space. Prove that $\mathcal{L}(\mathbb{V}, \mathbb{R})$ is a finite-dimensional vector space by explicitly constructing a basis for $\mathcal{L}(\mathbb{V}, \mathbb{R})$. Further, prove that $\dim\, \mathcal{L}(\mathbb{V}, \mathbb{R}) = \dim\, \mathbb{V}$.*

PROBLEM 8. *Let $T \in \mathcal{L}(\mathbb{V})$. Prove that*

$$T \text{ isomorphism} \iff \mathcal{N}(T) = \{0\} \iff \mathcal{R}(T) = \mathbb{V}.$$

PROBLEM 9. *Let $(\mathbb{V}, \|\cdot\|)$ be a normed vector space and let $T \in \mathcal{L}(\mathbb{V})$ be such that $\|T(v)\| \leqslant \|v\|$ for all $v \in \mathbb{V}$. Prove that $T - 2I$ is invertible, where I is the identity map on $\mathbb{V}$.*

PROBLEM 10. *Let $\mathbb{V}$ and $\mathbb{W}$ be two finite-dimensional spaces. Prove that $\mathbb{V}$ is isomorphic to $\mathbb{W}$ if and only if $\dim \mathbb{V} = \dim \mathbb{W}$.*

PROBLEM 11. *Suppose $\mathbb{V}$ is finite-dimensional and $T_1, T_2 \in \mathcal{L}(\mathbb{V})$. Prove that*

$$\dim\, \mathcal{N}(T_1 \circ T_2) \leqslant \dim\, \mathcal{N}(T_1) + \dim\, \mathcal{N}(T_2).$$

PROBLEM 12. *Suppose $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$ and $T' \in \mathcal{L}(\mathbb{W}, \mathbb{V})$ are such that $T \circ T' = \mathrm{Id}$. Show that $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T) = \mathbb{W}$.*

PROBLEM 13. *Let $T \in \mathcal{L}(\mathbb{V}, \mathbb{W})$, where $\mathbb{V}$ and $\mathbb{W}$ are finite-dimensional spaces. Define*

$$S : \mathcal{R}(T^*) \to \mathcal{R}(T), \quad \forall\, x \in \mathcal{R}(T^*) : \; Sx = Tx.$$

Prove that S is a linear isomorphism. Conclude that $\operatorname{rank}(T^*) = \operatorname{rank}(T)$.

PROBLEM 14. *Recall that we use $A \sim B$ to mean that A and B are similar. Show that $\sim$ is an equivalence relation on the set $\mathbb{F}^{n\times n}$, i.e., $\sim$ is reflexive, symmetric, and transitive.*

PROBLEM 15. *Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ be such that the equation $Ax = b$ admits a solution. Prove that there exists exactly one solution x of the form $x = A^\top z$ for some $z \in \mathbb{R}^m$.*

PROBLEM 16. *Let $\mathcal{G}$ be an undirected graph with vertices $\mathcal{V} = \{1, \dots, n\}$ and edges $\mathcal{E}$. Define $L \in \mathbb{R}^{n\times n}$ as follows:*

$$L_{ij} = \begin{cases} d_i & \text{if } i = j, \\ -1 & \text{if } (i,j) \in \mathcal{E}, \\ 0 & \text{otherwise,} \end{cases}$$

where d_i is the degree of vertex i.

(1) *Show that L is PSD but not PD.*
(2) *Prove that* $\operatorname{nullity}(L) = 1$ *if and only if $\mathcal{G}$ is connected. More generally, prove that* $\operatorname{nullity}(L)$ *is the number of connected components of $\mathcal{G}$.*

PROBLEM 17. *Let $A, B \in \mathbb{C}^{m\times n}$. Show that*

$$\forall\, 1 \leqslant i \leqslant n : \quad \lambda_i(A) - \|B\|_2 \leqslant \lambda_i(A+B) \leqslant \lambda_i(A) + \|B\|_2.$$

In particular, explain how we can conclude that the map $A \mapsto \lambda_i(A)$ is continuous.

PROBLEM 18. *Let $\alpha \neq \beta$ and $A \in \mathbb{C}^{n\times n}$. Establish from first principles that*

$$\mathcal{N}\big((A - \alpha I_n)(A - \beta I_n)\big) = \mathcal{N}(A - \alpha I_n) \oplus \mathcal{N}(A - \beta I_n).$$

Using the above decomposition, prove that A is similar to

$$\begin{pmatrix} \alpha I_k & 0 \\ 0 & \beta I_{n-k} \end{pmatrix}$$

if and only if $(A - \alpha I_n)(A - \beta I_n) = 0$. ($I_p$ denotes the identity matrix on $\mathbb{R}^{p\times p}$).

PROBLEM 19. *Let $\mathbb{V}$ be a finite-dimensional space. Let $P \in \mathcal{L}(\mathbb{V})$ be a projection operator on $\mathbb{V}$. Let $[P]$ be a matrix representation of P with respect to some basis of $\mathbb{V}$. Prove that rank P = trace $[P]$.*

PROBLEM 20. *For any $A \in \mathbb{R}^{n\times n}$, prove that there exists an orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ and a PSD matrix $P \in \mathbb{R}^{n\times n}$ such that $A = QP$.*

PROBLEM 21. *Let $P_1, \dots, P_m \in \mathcal{L}(\mathbb{V})$ be projection operators such that $P_1 + \cdots + P_m$ is the identity operator on $\mathbb{V}$. Prove that*

(1) $\operatorname{rank} P_1 + \cdots + \operatorname{rank} P_m = \dim \mathbb{V}$, *and*
(2) $\mathbb{V} = \mathcal{R}(P_1) \oplus \cdots \oplus \mathcal{R}(P_m)$.

PROBLEM 22. *For $A, B \in \mathbb{R}^{n\times n}$, define its Hadamard product $A \circ B \in \mathbb{R}^{n\times n}$ to be*

$$\forall\, 1 \leqslant i, j \leqslant n: \quad (A \circ B)_{ij} = A_{ij}B_{ij}.$$

Let $\|A\|_2$ be the spectral norm of A and $\lambda_{\max}(B)$ be the largest eigenvalue of B. Prove that

(1) $\forall\, C, D \in \mathbb{S}^n_+ :\ \|C \circ D\|_2 \leqslant \lambda_{\max}(C)\lambda_{\max}(D)$.
(2) $\forall\, A, B \in \mathbb{R}^{n\times n} :\ \|A \circ B\|_2^2 \leqslant \|(A^\top A) \circ (B^\top B)\|_2$.

Using these, prove that $\|A \circ B\|_2 \leqslant \|A\|_2 \|B\|_2$ for all $A, B \in \mathbb{R}^{n\times n}$.

PROBLEM 23. *Let $A, B \in \mathbb{C}^{n\times n}$. Prove that*

(1) *$I - AB$ invertible $\iff$ $I - BA$ invertible.*
(2) *λ is an eigenvalue of AB $\iff$ λ is an eigenvalues of BA.*

PROBLEM 24. *Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric matrices and let A be positive definite. Prove that the eigenvalues of $A^{-1}B$ are real. Moreover, if B is positive definite, then prove that the eigenvalues of $A^{-1}B$ are positive.*

PROBLEM 25. *Let $A \in \mathbb{C}^{n\times n}$ and $\tau > 0$ be such that*

$$\forall\, 1 \leqslant i \leqslant n: \quad \sum_{j=1}^{n} |A_{ij}| \leqslant \tau.$$

Prove that $\rho(A) \leqslant \tau$.

PROBLEM 26. *Suppose $A \in \mathbb{R}^{n\times n}$ is diagonalizable and two of its eigenvalues are equal. Show that $v, Av, \ldots, A^{n-1}v$ are linearly dependent for all $v \in \mathbb{R}^n$.*

PROBLEM 27. *The spectral norm of $A \in \mathbb{R}^{n\times n}$ is defined as*

$$\|A\|_2 = \max\left\{\|Ax\| :\ \|x\| = 1\right\},$$

where $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^n$. Prove that

$$\|A\|_2 = \begin{cases} \max\left\{\|Ax\| : \|x\| \leqslant 1\right\}. \\ \max\left\{y^\top Ax : \|x\| \leqslant 1 \text{ and } \|y\| \leqslant 1\right\}. \\ \sigma_{\max}(A). \end{cases}$$

PROBLEM 28. *Suppose $u, v \in \mathbb{V}$, where $\mathbb{V}$ is a real inner-product space. Prove that $\langle u, v\rangle = 0$ if and only if $\|u\| \leqslant \|u + cv\|$ for all $c \in \mathbb{R}$.*

PROBLEM 29. *Let $\mathbb{V}$ be a real inner-product space. Let $f : \mathbb{V} \to \mathbb{V}$ be such that*

$$\forall\, x, y \in \mathbb{V}: \quad \langle f(x), f(y)\rangle = \langle x, y\rangle.$$

Prove that f is linear. On the other hand, give a simple example to show that if

$$\forall\, x \in \mathbb{V}: \quad \langle f(x), f(x)\rangle = \langle x, x\rangle,$$

then f need not be linear.

PROBLEM 30. *For some $A \in \mathbb{C}^{n\times n}$, suppose $x^{\mathrm{H}}Ax = 0$ for all $x \in \mathbb{C}^n$. Show that $A = 0$. On the other hand, give an example of $A \in \mathbb{R}^{2\times 2}$ s.t. $A \neq 0$ but $x^\top Ax = 0$ for all $x \in \mathbb{R}^2$.*

PROBLEM 31. *Let A and B be PSD. Prove that $A^2 = B^2$ only if $A = B$.*

PROBLEM 32. *Using the permutation-based formula for determinants, prove that $\det(AB) = \det(A)\det(B)$ for any $A, B \in \mathbb{R}^{n\times n}$.*